

Eigenmode approximations of noisy tipping-point systems



Felix Shaw-Bell

Department of Mathematics
The University of Auckland

Supervisor: Professor Graham Donovan

A thesis submitted in partial fulfilment of the requirements for the degree of MSc in Applied
Mathematics, The University of Auckland, 2026.

Abstract

Tipping point systems are characterised by large, abrupt, persistent departures from some stable state, often following some relatively small change in one or more underlying parameter(s). There are many real-world tipping point systems in ecology and climatic science for which tipping events can have significant, and potentially irreversible, negative impacts. Modelling of the risk and timing of tipping events is of high utility given the negative impacts. A simple model for such systems is a one dimensional stochastic differential equation (SDE) which undergoes a loss of stability (bifurcation) in the deterministic dynamics. The Fokker-Planck equation (FPE) describes the temporal evolution of the probability density function (PDF) of such an SDE. Although methods in tipping point research have been previously applied to approximate the FPE, there remains incomplete coverage of settings which are well approximated. One such existing method used to approximate the FPE is an eigenmode approximation, whereby the evolution of a PDF governed by the FPE is approximated with the eigenvalues and eigenfunction of the Fokker-Planck operator (FPO). The eigenvalues and eigenfunctions of the FPO are not generally available in closed form and typically require approximations. We study and expand upon the boundary condition perturbation method (BCPM) and compare it with the Rayleigh-Schrödinger perturbation method (RSPM), a classical method in quantum mechanics, to approximate the eigenvalues and eigenfunctions of the FPO. Each method offers accurate approximations in different settings, with the BCPM performing best in low noise settings, and the RSPM performing best in high noise settings. In the time-dependent setting, the BCPM performs well to approximate the tipping rates given a slow evolution of the system dynamics and low noise. In comparison, the RSPM performs best in high noise settings but often cannot track the evolution of a given PDF.

Contents

Abstract	i
1 Introduction	1
2 Literature review	5
3 Preliminaries	11
3.1 Fokker-Planck equation	11
3.2 General results from linear algebra	14
3.3 Sturm-Liouville theory	14
3.4 Schrödinger operator	15
3.5 Common transformations	17
4 Eigenmode expansion	23
4.1 General framework	23
4.2 Adiabatic regime	29
4.3 Non-adiabatic regime	31
5 Eigenvalues and eigenfunctions of the Fokker-Planck operator	33
5.1 Eigenvalues and eigenfunctions of solvable potential functions	33
5.2 Boundary condition perturbation method	39
5.3 Rayleigh–Schrödinger perturbation method	48
6 Results	55
6.1 Experimental setting	55
6.2 Approximations of eigenvalues and eigenfunctions	58
6.3 Empirical tipping rates	61
6.4 Eigenmode approximations	64
7 Discussion	69

8 Conclusion	71
References	71

List of Figures

1.1	Examples of types of tipping effects.	3
1.2	Example of system with both B-tipping and N-tipping effects.	4
2.1	Potential functions with one and two escape barriers.	9
3.1	Examples of boundary conditions represented in potential functions.	13
3.2	Effect of moving basin of attraction.	19
5.1	Examples of known eigenfunctions.	40
5.2	BCPM approximation applied to an asymmetric potential function.	44
5.3	BCPM approximation for an even anti-confining potential function.	47
5.4	BCPM approximation for an even confining potential function.	49
6.1	Potential function used for experiments.	57
6.2	Convergence of RSPM.	59
6.3	Errors of leading non-zero eigenvalue approximations of RSPM and BCPM. . .	60
6.4	Leading non-zero eigenvalue best approximation method across parameter space.	61
6.5	Errors of higher non-zero eigenvalue approximations of the RSPM.	62
6.6	Example of an SDE realisation with a bistable potential function.	62
6.7	Errors of leading non-eigenvalue used as an estimation of empirical tipping rate.	63
6.8	Example of probability of being found in right hand potential well.	64
6.9	Probability of being found in right hand potential well across parameter space. .	65
6.10	Errors of 4-mode RSPM approximation.	66
6.11	Errors of 2-mode BCPM approximation.	67

Chapter 1

Introduction

Systems that undergo a large, abrupt, persistent departure from some stable state, often following some relatively small change in one or more underlying parameter(s), are called *tipping point systems*. An event in which such a change in state takes place is called a *tipping event* (also called a *critical transition*). There are several climatic and ecological systems that show (or are theorised to show) tipping phenomena. It is often the case that these abrupt changes in state are catastrophic, with negative consequences.

A relevant example of a tipping point system is the Atlantic Meridional Overturning Circulation (AMOC). AMOC is an important climatic system which transports warm water from the tropical Atlantic towards Northern Europe and returns cold arctic water towards the equator [19]. A change in the input of fresh water in the Northern Atlantic (amongst other parameters) is theorised to have the ability to collapse AMOC and potentially cause large, abrupt changes in the climates of not just Northern Europe, but much of the globe [30]. Palaeoclimatic data suggests that repeated climatic tipping events, potentially caused by past collapses of AMOC, have occurred in the North Atlantic over the past 120,000 years [6, 18]. Research published in 2024 suggests that AMOC is presently ‘on route to tipping’ [29].

There are several ecological systems that display tipping phenomena. Notable examples include, abrupt transitions between savanna and forest [27], collapse of over exploited fisheries [2] and abrupt transitions in lakes between a vegetation dominated state to an algal dominated state [24]. Table 1 of [7] provides a summary of ecological tipping systems along with potential evolutionary responses.

Given the significant potential impacts of real-world tipping point systems, the ability to model, and thus predict, where and when a tipping event might occur is of great importance. In the case of stochastic systems, the additional question of tipping probability must be considered.

Consider a one-dimensional, time dependant state variable $X(t)$, the evolution of which is

governed by the following stochastic differential equation (SDE),

$$dX = f(X, \mu(t)) dt + \sigma dW, \quad (1.1)$$

where f is some function providing deterministic dynamics (also called the *drift* or *drift function*), $W(t)$ is the standard Wiener process providing random additive noise (also called *diffusion*) and σ is the standard deviation of the noise. f is dependent on some underlying time-dependent parameter $\mu(t)$ (also called the *bifurcation parameter* in the case that μ causes a bifurcation, as will be the case in this thesis). We assume that there exists a region(s) for which f produces a *basin of attraction*. A basin of attraction is some region about a stable equilibrium (equilibrium of the drift function) $x^*(t)$, for which X is attracted towards x^* by the drift function f . More precisely, let x^* be defined by the following properties, $f(x^*, \mu) = 0$ (equilibrium point) and $f_x(x^*, \mu) < 0$ (stability), where $f_x = \frac{\partial f}{\partial x}$. As such, $x^*(t)$ may evolve over time as f evolves, and may even cease to exist following a bifurcation of f . In the context of this model, we consider a tipping event to be a large, abrupt and persistent departure of the state variable X from the stable equilibrium x^* (or from where the stable equilibrium used to be, in the case of a bifurcation).

Following the classification by Ashwin et al. [1], three categories of *tipping effects* can be considered for the model described by Eq. (1.1) (see Fig. 1.1 for an illustrative example of each tipping effect):

- (i) *B-tipping* in which changes in μ produce a bifurcation in the drift function f , resulting in the loss of stability of the basin of attraction.
- (ii) *N-tipping* in which a noisy excursion produced by W is large enough to cause X to exit the basin of attraction (that is, an excursion that passes over an unstable equilibrium).
- (iii) *R-tipping* in which changes in μ shift the location of the basin of attraction at a rate which is faster than X can adapt, causing X to exit the basin of attraction.

In the case of purely deterministic systems (that is, $\sigma = 0$ in Eq. (1.1)), B-tipping can be studied using classical bifurcation analysis. Purely deterministic R-tipping can often be studied with a moving frame of reference. In the case of a linear shift of the drift function (as shown in Fig. 1.1c), tipping points can be computed analytically, while non-linear shifts require more advanced techniques and approximations [1]. The case of N-tipping requires probabilistic methods to be utilised. In addition, many tipping point systems display more than one tipping effect acting at once. For example a noisy system ($\sigma > 0$) may approach a bifurcation, causing the basin of attraction to weaken, increasing the probability of N-tipping (potentially never actually undergoing a bifurcation). A similar effect may occur in noisy systems showing increasing rate of parameter change (see [22] for an example of N-tipping and R-tipping dynamics operating in tandem).

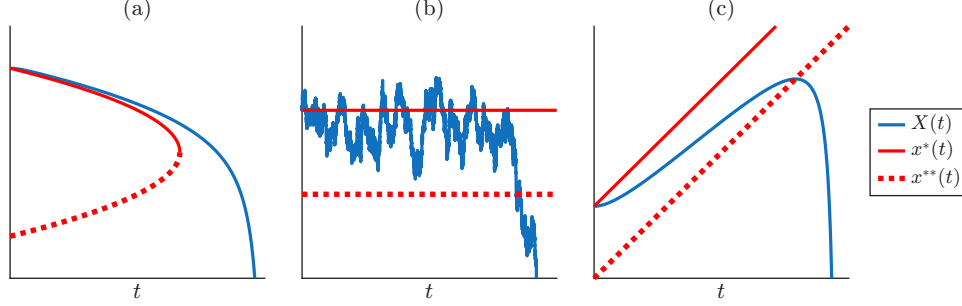


Figure 1.1: Examples of (a) B-tipping, (b) N-tipping and (c) R-tipping. In each case we observe an abrupt change in the state variable X following either changes in the existence/location of the stable equilibrium x^* and unstable equilibrium x^{**} of f , or due to a large noisy excursion. The changes in equilibrium can be attributed to the effects of a time-dependent underlying parameter $\mu(t)$ on $f(X, \mu)$.

In this thesis we consider noisy systems approaching a bifurcation of the drift function, that is, both N-tipping and B-tipping effects. See Fig. 1.2 for an example of such a system. In such a case, classical bifurcation analysis is not sufficient to describe the tipping dynamics at hand. Instead, the probabilistic behaviour of the system must be quantified. Ideally, the complete probability density function (PDF) of the state variable can be found. The time evolution of the PDF $p(x, t)$ of the state variable $X(t)$, as defined by Eq. (1.1), can be described by the *Fokker-Planck equation* (FPE) as such [20],

$$\frac{\partial p}{\partial t} = D \frac{\partial^2 p}{\partial x^2} - \frac{\partial}{\partial x} [fp], \quad (1.2)$$

where $D = \frac{\sigma^2}{2}$. We call D the *diffusion constant*. Note that $X(t)$ is a time-dependent random process whereas x is an independent spatial variable. It is often not possible to find solutions to the FPE in an analytic manner. As a result, approximations or simplifying assumptions must be developed. However, despite the importance of such solutions in the application of tipping point systems, general (non-numerical) methods that are effective over a wide range of system parameters are still not fully developed. This thesis aims to expand upon existing methods and apply new complementary methods to tipping point problems in an effort to better answer the question:

With what probability might a tipping point system of interest undergo a tipping event and how does this probability vary with time and choice of system parameters?

Specifically we wish to answer this question for a one-dimensional SDE with a time-dependent bifurcation parameter, as described by Eq. (1.1). This thesis tackles the problem

statement by utilising an *eigenmode approximation* of the PDF. The eigenvalues and eigenfunctions used in the eigenmode approximation are approximated by two perturbation methods, the *boundary condition perturbation method* (BCPM) and the *Rayleigh-Schrödinger perturbation method* (RSPM). The BCPM has already been applied to tipping point systems by Ritchie and Sieber [21, 22] for a certain class of drift functions in low noise settings. This thesis expands upon to the method to allow symmetric confining drift functions and symmetric anti-confining drift functions to be approximated under such a method. We also apply the RSPM to provide an alternative approximation method which performs well in high noise settings (for which the BCPM does not perform well).

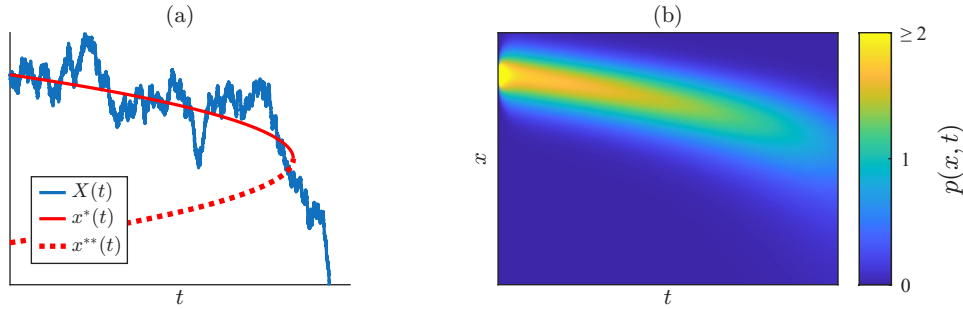


Figure 1.2: (a) A realisation X of a system with both B-tipping and N-tipping effects, including the locations of the stable equilibrium x^* and unstable equilibrium x^{**} , and (b) a numerically computed solution p to the corresponding FPE.

The thesis begins in Chap. 2 with a review of methods relevant to the study of the FPE in the context of tipping point systems. In Chap. 3 details of the FPE are presented, along with an overview of foundational mathematical techniques required for analysis of the FPE. In Chap. 4 the eigenmode approximation is presented, along with supporting analysis. Exact and approximated eigenvalues and eigenfunctions are then presented in Chap. 5. In this chapter the BCPM and the RSPM are presented for eigenvalue and eigenfunction approximation. Results showing a comparison of the performance of each perturbation method (for a variety of system parameters) are then presented in Chap. 6. Chap. 7 discusses the results and methods used and suggests future research. Finally, concluding remarks are made in Chap. 8.

Chapter 2

Literature review

We now review several existing methods that have made contributions to addressing the problem statement. Some of the following methods contributed directly to the work in this thesis, while others act as context for the problem at hand.

Kuehn's fast-slow regime

A common framework under which to study tipping points as modelled by Eqs. (1.1) and (1.2) is *Kuehn's fast-slow regime* [16]. Under the fast-slow regime we consider a slowly changing parameter $\mu(t)$ (relative to fast changes in $X(t)$). We let $\mu(t)$ change in a slow, linear manner,

$$d\mu = \varepsilon dt,$$

where $0 < \varepsilon \ll 1$ is some small, positive parameter. Let us call t the *fast* timescale and $\tau := \varepsilon t$ the *slow* timescale. That is, fluctuations of X (and thus the evolution of p) are observed on the fast timescale while changes to μ occur on the slow timescale.

Under the fast-slow regime, close to a given t it is often reasonable to assume that $\mu(t)$ is approximately constant. It is also often reasonable to assume that following some small increase along the slow timescale τ , the PDF p of X has had sufficient time on the fast timescale t to relax to approximately the steady state distribution.

A fast-slow regime is a realistic regime under which to study many tipping point systems. It is often the case in real-world tipping systems that the time scale at which the state variable X fluctuates is much faster than that at which the underlying parameter μ changes. Consider for example climate systems such as AMOC. Weather patterns come and go on a scale of days or weeks (fast time scale) while parameters such as atmospheric carbon-dioxide concentration or rates of ice sheet melt change on a scale of years or decades (slow time scale).

Critical slowing down

When studying a real world system, the governing equations may not be exactly known. As a result, one must find *early warning signals* (EWS) from observable state variables. A well known phenomena which is often considered when constructing an EWS in tipping point systems is *critical slowing down* (CSD).

To informally motivate CSD, consider a noisy system modelled by Eq. (1.1) approaching a bifurcation (due to the bifurcation parameter $\mu(t)$), eventually causing the loss of the stable equilibrium x^* . In a one dimensional system, such a bifurcation occurs at $f_x(x^*, \mu(t)) = 0$. As x^* is a stable equilibrium prior to the bifurcation, we know $f_x(x^*, \mu(t)) < 0$ prior to the bifurcation. Therefore, (assuming continuity and general well behaved properties) we expect $f_x(x^*, \mu(t))$ to approach 0 as a bifurcation approaches, resulting in weakening drift towards x^* and a ‘shallower’ basin of attraction.

When observing a realisation $X(t)$ of such a system approaching a bifurcation, we typically see three effects; The characteristic return time to the equilibrium increases, the variance of a $X(t)$ increases, and the auto-correlation of $X(t)$ increases [8, 31]. These three effects are observable variables that arise from CSD which can each be used as an EWS (i.e. observing increasing variance indicates an impending bifurcation). Several of the following methods provide approximations of the FPE with the intention of estimating variance.

Kuehn’s quasi-steady state approximation

Kuehn’s quasi-steady state approximation is a method for approximating a steady state solution $\hat{p}_\infty(x)$ of Eq. (1.2) with a drift function $f(x, \mu)$ for a given μ within some basin of attraction [16]. The method proceeds by restricting the domain to only the basin of attraction and applying reflecting boundary conditions at the boundaries of the basin of attraction. The method assumes that we are in Kuehn’s fast-slow regime. The boundaries of the basin of attraction are given by the unstable equilibrium of f , or $\pm\infty$ in the case that no unstable equilibrium exists on a given boundary of the basin of attraction. Let the basin of attraction, and thus our restricted domain, be $x \in [a, b]$ (where a or b are either locations of unstable equilibrium or $\pm\infty$). With reflecting boundary conditions at each boundary of the basin of attraction, it is guaranteed that a steady state solution exists. After restricting the domain, we solve for $\hat{p}_\infty(x)$ by assuming Eq. (1.2) is in steady state as such,

$$D \frac{d^2 \hat{p}_\infty}{dx^2} - \frac{d}{dx} [f \hat{p}_\infty] = 0,$$

Solving with the reflecting boundary conditions gives,

$$\hat{p}_\infty = C e^{-U/D},$$

where $U = -\int f dx$ and C is a constant of integration which can be solved for with the normalisation condition $\int_a^b \hat{p}_\infty(x) dx = 1$.

The variance $\text{Var}(\hat{p}_\infty)$ can be computed and used as an approximation of the variance of the state variable $X(t)$ for a given time t ,

$$\text{Var}(\hat{p}_\infty) = \int_a^b x^2 \hat{p}_\infty(x) dx - \left(\int_a^b x \hat{p}_\infty(x) dx \right)^2.$$

Linearisation to the Ornstein–Uhlenbeck process

A standard tool used to study the dynamics of $X(t)$ near x^* is *linearisation* of Eq. (1.1) about the stable equilibrium x^* [23]. That is, we assume that a drift function $f(x, \mu)$ for a given μ is well approximated by a first order Taylor series (in x) about x^* . The resulting linearised state variable $\hat{X}$ is governed by the following SDE,

$$d\hat{X} = f_x(x^*, \mu)(\hat{X} - x^*) dt + \sigma dW, \quad (2.1)$$

where once again $f_x = \frac{\partial f}{\partial x}$. Eq. (2.1) is the SDE that defines the *Ornstein–Uhlenbeck* (OU) process. As such, this approximation method will be referred to as *OU linearisation*. The resulting FPE describing the time evolution of the PDF $\hat{p}(x, t)$ of $\hat{X}$ is,

$$\frac{\partial \hat{p}}{\partial t} = D \frac{\partial^2 \hat{p}}{\partial x^2} - \frac{\partial}{\partial x} [f_x(x^*, \mu)(x - x^*) \hat{p}], \quad (2.2)$$

where $D = \frac{\sigma^2}{2}$. The steady state solution $\hat{p}_\infty(x)$ (different $\hat{p}_\infty$ presented prior in Kuehn's quasi-steady state) of Eq. (2.2) for a given drift function f can be calculated to be,

$$\hat{p}_\infty(x) = \sqrt{\frac{\alpha}{2\pi D}} e^{-\alpha(x-x^*)^2/2D}, \quad (2.3)$$

where $\alpha = f_x(x^*, \mu)$. Eq. (2.3) is the Gaussian distribution with mean $\mathbb{E}(\hat{p}_\infty) = x^*$ and variance $\text{Var}(\hat{p}_\infty) = \frac{D}{\alpha}$. Therefore, OU linearisation produces an analytical approximation of the variance of the state variable X .

Donovan's distribution moment approximation

Another method for approximating the variance (and higher moments if desired) of the state variable X is presented by Donovan [9]. *Donovan's distribution moment approximation* finds an approximation for the k th moment M_k ($k = 0, 1, 2, \dots$) of the PDF p (as described by the FPE in Eq. (1.2)). The method operates under Kuehn's fast-slow regime (and the assumptions which follow) and utilises properties of polynomial type drift functions $f(x, \mu)$, that is we assume we can express f as,

$$f(x, \mu) = \sum_i c_i x^{n_i} \mu^{m_i},$$

for some real constants c_i and integers n_i and m_i . The raw moments for a given μ are given by

$$M_k = \int_{-\infty}^{\infty} x^k p \, dx.$$

Following some closure assumption (i.e. Gaussian assumption) one can approximate a finite number of moments M_k . Under a Gaussian closure assumption the mean and variance can be approximated.

Kramers' escape rate

Methods mentioned up to this point have provided approximations of the PDF described by Eq. (1.2). However, these methods do not make an attempt to quantify the probability/dynamics of tipping over the unstable equilibrium. Kuehn's quasi-steady state and OU linearisation do not feature the unstable equilibrium over which X might tip. Kuehn's quasi-steady removes the unstable equilibrium by restricting the domain of the system to the only the basin of attraction and OU linearisation has no unstable as it assumes the system is well approximated by the OU process. The probability of tipping is a key metric which should be approximated.

It is often useful to consider the *potential function* (also called simply the *potential*) $U(x, \mu)$, defined as $U = - \int f \, dx$, when describing the deterministic dynamics of a given tipping point system. The potential function U can be thought of as a 'physical landscape' with gravity acting downwards as a representation of the drift function f . That is, the base of a valley in U corresponds to a stable equilibrium in f and a crest in U corresponds to an unstable equilibrium in f . When referring to potential functions, the basin of attraction near a stable equilibrium is called a *potential well* (which visually looks like a basin or well when U is plotted).

We call an *escape barrier* a region between a stable equilibrium and an unstable equilibrium which must be overcome by noise for a realisation X to escape a potential well. For example, in Fig. 2.1a there is a single escape barrier of magnitude ΔU between the stable equilibrium x^* and the unstable equilibrium x^{**} , while in Fig. 2.1b there are two escape barriers, one of magnitude ΔU_L between x_L^{**} and x^* and another of magnitude ΔU_R between x^* and x_R^{**} .

The *escape rate* r (also called *tipping rate* and *transition rate*) is the probability of a realisation $X(t)$ of Eq. (1.1) escaping from the potential well per unit time. A classical method for estimating the rate of escape of a noisy particle from a potential well is given by Kramers [15,20]. Originally motivated by reaction-rate theory in the field of chemistry, *Kramers' escape rate* r_K approximates the escape rate of Eq. (1.1), under the assumptions that the potential well is deep relative to the diffusion constant D (that is, the deterministic forces dominate the diffusive forces). This assumption allows integrals of the PDF $p(x, t)$ to be well approximated by an integral over a small region near $x^*(t)$ following a linearisation approximation (in a similar manner to the OU linearisation method).

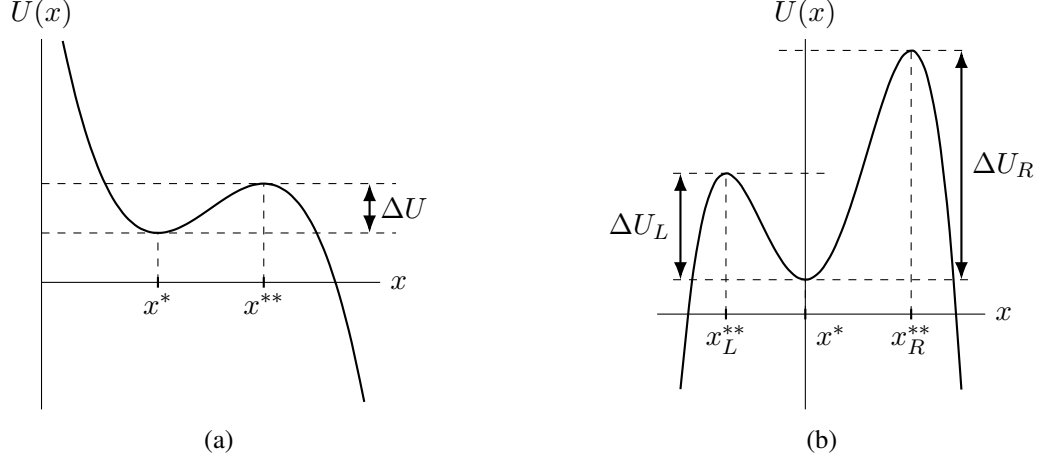


Figure 2.1: (a) A potential well with a single escape barrier and (b) a potential well with two escape barriers.

The resulting Kramers' escape rate in the case of a single escape barrier (see Fig. 2.1a) is given by,

$$r_K = \frac{1}{2\pi} \sqrt{-U_{xx}(x^*)U_{xx}(x^{**})} e^{-\Delta U/D}, \quad (2.4)$$

where $\Delta U = U(x^{**}) - U(x^*)$ and $U_{xx} = \frac{\partial^2 U}{\partial x^2}$. A similar treatment can be applied to the case of a potential well with an escape barrier on either side (see Fig. 2.1b). The resulting approximation for the total escape rate r_K from the potential well is simply the sum the escape rate over each barrier as such,

$$r_K = r_{KL} + r_{KR},$$

where r_{KL} and r_{KR} are the Kramers' escape rates given by Eq. (2.4) over the left and right barriers, respectively.

Although Kramers' escape rate is only accurate given that the potential well is deep relative to the diffusion constant, it still provides a good general scale for the relationship between the strength diffusivity, the depth of potential well and the escape rate. That is, we can very approximately say $r \propto e^{-\Delta U/D}$.

Eigenmode approximation

A method of key importance for this thesis is the *eigenmode approximation* of solutions to the FPE, as presented by Ritchie and Sieber [21, 22]. The eigenmode approximation will be covered in detail in the following chapters, the general approach is applied as follows; For most relevant examples of the drift function f , the eigenfunctions of the Fokker-Planck operator

form an orthonormal basis and can thus be used to describe the PDF p . Ritchie and Sieber in particular utilise Kuehn's fast-slow regime and an assumption of a strong basin of attraction relative to the level of noise to truncate the eigenmode approximation to include only the leading eigenfunction. Under such assumptions, the leading non-zero eigenvalue well approximates the escape rate of X from the basin of attraction.

Chapter 3

Preliminaries

This chapter presents the terminology, known results and methods that are to be used throughout the remainder of the thesis. The FPE is restated in more detail and classical supporting results are presented in Sec. 3.1. In Sec. 3.2 we give several known results from linear algebra. The basic results from Sturm-Liouville theory are stated in Sec. 3.3. Common transformations of the FPE are then presented. Notably, the transformation and equivalence between the eigenvalue problem of the Fokker-Planck operator and the eigenvalue problem of the Schrödinger operator is covered in Sec. 3.4, followed by other common transformations in Sec. 3.5.

3.1 Fokker-Planck equation

The FPE is central to the analysis throughout the entirety of this thesis. It provides the tools we require to analyse tipping point systems modelled by a stochastic differential equation (SDE). In Chap. 1 the one dimensional SDE model was presented, along with the corresponding FPE. We restate each now in more detail.

Let us consider the stochastic process $(X(t))_{t \geq 0}$ which is described by the following SDE,

$$\begin{cases} dX(t) = f(X(t), \mu(t)) dt + \sigma dW(t), \\ X(0) = x_0, \end{cases} \quad (3.1)$$

where the *drift function* $f(x, \mu(t))$ represents the deterministic drift, $\sigma > 0$ is the standard deviation of the additive noise, $(W(t))_{t \geq 0}$ is the standard Wiener process, $\mu(t)$ is a time-dependent parameter, and x_0 is the initial condition of $X(t)$.

Let us introduce $p(x, t)$ as the probability density function (PDF) of $X(t)$. The time-evolution of p is described by the FPE as follows [11, 20],

$$\frac{\partial p}{\partial t} = D \frac{\partial^2 p}{\partial x^2} + \frac{\partial}{\partial x} \left[\frac{\partial U}{\partial x} p \right], \quad (3.2)$$

where $\frac{\partial U}{\partial x} = -f$ and $D = \frac{\sigma^2}{2}$ is the *diffusion constant*. $U(x, \mu(t))$ is the potential function of our stochastic system. In many cases U will be a more convenient means of expressing the deterministic dynamics in our system than f . The initial condition $X(0) = x_0$ for X given in Eq. (3.1) has an equivalent initial condition for the PDF p , stated in terms of the Dirac delta function δ as such,

$$p(x, 0) = \delta(x - x_0). \quad (3.3)$$

When working with the PDF p we may wish to provide an initial distribution p_0 , rather than an initial value, as such,

$$p(x, 0) = p_0(x). \quad (3.4)$$

We now introduce the operator notation for the FPE. We define the *Fokker-Planck operator* (FPO) $\mathcal{L}_F$, acting on some function p , as such,

$$\mathcal{L}_F p = D \frac{\partial^2 p}{\partial x^2} + \frac{\partial}{\partial x} \left[\frac{\partial U}{\partial x} p \right]. \quad (3.5)$$

That is, we can write Eq. (3.2) in terms of $\mathcal{L}_F$,

$$\frac{\partial p}{\partial t} = \mathcal{L}_F p. \quad (3.6)$$

One should also note that due to the time dependence of $U(x, \mu(t))$, $\mathcal{L}_F$ itself is dependent on time. The *probability flux* $J(x, t)$ quantifies the (net) flow of probability through the PDF $p(x, t)$ at a given time and location. Probability flux is defined as [20],

$$\frac{\partial p}{\partial t} = -\frac{\partial J}{\partial x},$$

that is,

$$J(x, t) = -D \frac{\partial p}{\partial x} - \frac{\partial U}{\partial x} p. \quad (3.7)$$

When considering a potential function U for a given μ , we call the potential function *confining* if $U(x, \mu) \rightarrow +\infty$ as $|x| \rightarrow \pm\infty$. We call the potential function *anti-confining* if $U(x, \mu) \rightarrow -\infty$ as $|x| \rightarrow \pm\infty$. When defining boundary conditions of $p(x, t)$ (or $\mathcal{L}_F p$) at some point $x = a$, there are two natural choices which will be considered in this thesis:

- An *absorbing* boundary condition is defined by $p(a, t) = 0$. As the name suggests, this boundary condition corresponds with probability density being absorbed or removed from the system at $x = a$.
- A *reflecting* boundary condition is defined by $J(a, t) = 0$. A reflecting boundary condition conserves probability density, that is, no probability density may enter or exit the system at $x = a$.

The effect of each boundary condition can be visualised (and is equivalent) with a potential function U . An absorbing boundary condition at $x = a$ has the effect of setting $U(a, \mu) = -\infty$, while a reflecting boundary condition has the effect of setting $U(a, \mu) = \infty$ (see [20] for more details).

It is often the case that analysis of an infinite domain must be carried out on a finite domain while retaining the correct qualitative behaviour (for example when applying numerical methods). As such, there is a natural choice of boundary conditions when restricting an infinite domain $x \in \mathbb{R}$ to a finite domain $x \in [a, b]$. If $U \rightarrow -\infty$ as $x \rightarrow \pm\infty$ it is natural to apply absorbing boundary conditions (i.e. set $U(a, \mu) = -\infty$), while if $U \rightarrow \infty$ as $x \rightarrow \pm\infty$ it is natural to apply reflecting boundary conditions (i.e. set $U(a, \mu) = \infty$). See Fig. 3.1 for illustrations of natural selection of boundary conditions applied to domain restrictions.

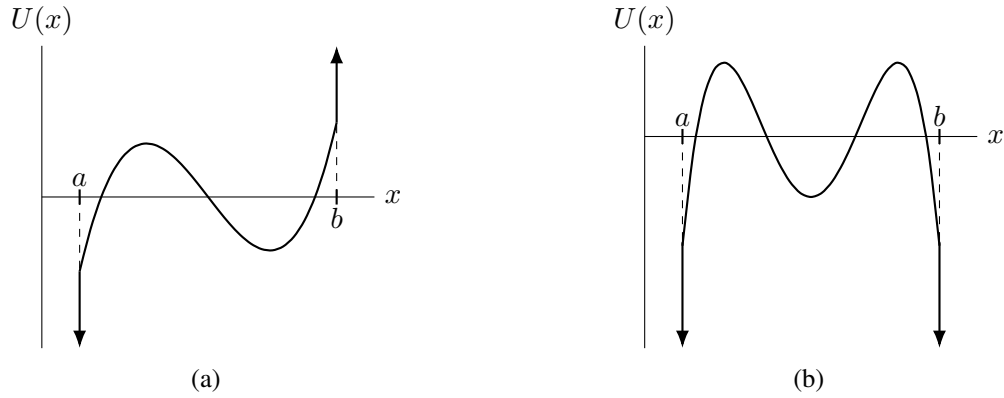


Figure 3.1: (a) Potential function with absorbing boundary condition at $x = a$ and reflecting boundary condition at $x = b$. (b) Potential function with absorbing boundary conditions at both $x = a$ and $x = b$.

It may also be the case that we wish to study the FPE with boundary conditions to model specific transient or survival behaviour of the system. For example, it may be of interest to know the probability of the random process $X(t)$ remaining entirely within the interval $[a, b]$ over time. We call such a probability the *survival probability* $S(t)$ (survival probabilities are not just applicable to the probability of remaining within an interval, other criteria can also be described by a survival probability). This can be achieved by finding the solution $p(x, t)$ of the FPE with two absorbing boundary conditions at $x = a$ and $x = b$. The survival probability is then given by,

$$S(t) = \int_a^b p(x, t) dx.$$

3.2 General results from linear algebra

A brief summary of important and relevant results from linear algebra will now be presented (see [28]). Although almost all the following results can be applied generally to both finite- and infinite-dimensional vector/function spaces, we will look to focus on the infinite-dimensional function spaces, given the relevance to the FPE.

Let $\mathcal{H}$ be an infinite-dimensional Hilbert space over a field F equipped with the norm $\|\cdot\|$ and inner product $\langle \cdot, \cdot \rangle$. Let $L : \mathcal{H} \rightarrow \mathcal{H}$ be a linear operator. For some $\psi \in \mathcal{H}$ and $\lambda \in \mathbb{C}$. If

$$L\psi = \lambda\psi, \quad (3.8)$$

then we call λ an *eigenvalue* of L and ψ the (corresponding) *eigenfunction* of L . Typically the eigenvalues and eigenfunctions will be indexed and given in sets, i.e. $\{\lambda_0, \lambda_1, \dots\}$ and $\{\psi_0, \psi_1, \dots\}$. The term *zero eigenvalue* refers to an eigenvalue equal to zero, $\lambda = 0$ and the *zero eigenfunction* refers to the corresponding eigenfunction. It will also be common for us to refer to *non-zero eigenvalues* and *non-zero eigenfunctions*. This should not be confused with labelling conventions, i.e. the zeroth (following some ordering of eigenvalues) eigenvalue λ_0 is not necessarily a zero-eigenvalue unless $\lambda_0 = 0$. As stated by Risken and Frank [20], a FPO has a zero-eigenvalue if and only if the system has two reflecting boundary conditions (or a confining potential function, which implicitly defines two reflecting boundary conditions).

In the case that L is time dependent, we call the eigenvalues and eigenfunctions for L at a given time the *instantaneous eigenvalues* and *instantaneous eigenfunctions*, respectively.

Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be Hilbert spaces. We define the *Hermitian adjoint* (or *adjoint*) $L^* : \mathcal{H}_2 \rightarrow \mathcal{H}_1$ of some operator $L : \mathcal{H}_1 \rightarrow \mathcal{H}_2$ as the operator satisfying the following condition,

$$\langle Lf, g \rangle = \langle f, L^*g \rangle \quad \forall f \in \mathcal{H}_1, g \in \mathcal{H}_2.$$

For the some Hilbert space $\mathcal{H}$, an operator $L : \mathcal{H} \rightarrow \mathcal{H}$ is a *self-adjoint* operator if $L = L^*$, or alternatively if,

$$\langle Lf, g \rangle = \langle f, Lg \rangle \quad \forall f, g \in \mathcal{H}. \quad (3.9)$$

Useful properties a self adjoint linear operator L include:

- (i) Every eigenvalue λ of L is real.
- (ii) Every eigenfunction ψ of L is orthogonal to every other eigenfunction, given that they correspond to distinct eigenvalues.

3.3 Sturm-Liouville theory

We now recall the important classical results of Sturm-Liouville theory (see [4]). A *Sturm-Liouville problem* is an ordinary differential equation of the following form,

$$\frac{d}{dx} \left[u(x) \frac{d\psi}{dx} \right] + v(x)\psi = -\lambda w(x)\psi, \quad (3.10)$$

for given functions $u(x)$, $v(x)$, and $w(x)$, on the domain $[a, b]$ along with some boundary conditions. Values of λ which yield non-trivial solutions are called the *eigenvalues* of the problem. The solution ψ associated with a given eigenvalue is called the *eigenfunction*. This terminology is analogous to linear operator definitions of eigenvalues and eigenfunctions and a Sturm-Liouville problem can in fact be written in terms of a linear operator. A Sturm-Liouville problem is said to be *regular* if the following conditions are satisfied,

- (i) The functions u , v , w and $\frac{du}{dx}$ are all continuous on $[a, b]$.
- (ii) $u(x) > 0$ and $w(x) > 0$ for all $x \in [a, b]$.
- (iii) The problem has boundary conditions of the form

$$\begin{cases} \alpha_1 \psi(a) + \alpha_2 \psi'(a) = 0, & \alpha_1, \alpha_2 \text{ not both } 0, \\ \beta_1 \psi(b) + \beta_2 \psi'(b) = 0, & \beta_1, \beta_2 \text{ not both } 0. \end{cases} \quad (3.11)$$

For any regular Sturm-Liouville problem, the following properties hold:

- (i) The eigenvalues $\{\lambda_0, \lambda_1, \dots\}$ are real, distinct and can be ordered such that $\lambda_0 < \lambda_1 < \dots \rightarrow \infty$. Furthermore, if $v(x) \geq 0$ for all $x \in [a, b]$ then $0 \leq \lambda_0 < \lambda_1 < \dots \rightarrow \infty$, that is, the eigenvalues are non-negative.
- (ii) Each eigenvalue λ_n has a unique (up to constant multiple) corresponding eigenfunction $\psi_n(x)$ called the *n th fundamental solution*.
- (iii) The set of normalised eigenfunctions $\{\psi_0, \psi_1, \dots\}$ form an orthonormal basis under the $w(x)$ weighted Hilbert space $L^2([a, b], w(x) dx)$.
- (iv) The n th eigenfunction ψ_n has exactly n zeros in the open interval (a, b) . This property is called the *Sturm oscillation theorem*.

3.4 Schrödinger operator

An important method for studying the FPO is analysis of its corresponding one dimensional, time-independent *Schrödinger operator* $\mathcal{L}_S$. Relevant results are recalled now (see [20]). We define the linear differential operator $\mathcal{L}_S$ acting on some function q as,

$$\mathcal{L}_S q = D \frac{d^2 q}{dx^2} - V(x)q, \quad (3.12)$$

where $V(x)$ is the potential function and D is the diffusion constant. Note that for the sake of consistency within this thesis, Eq. (3.12) differs from the conventional quantum mechanics definition of the Schrodinger operator by sign and organisation of constants (typically physicists write $D = \hbar/2m$ where $\hbar$ is the reduced Plank constant and m is the mass of a particle).

Let us consider the FPO operator for a given μ . We will therefore drop the μ notation in $U(x, \mu)$ for the time being. Let the eigenvalues of the FPO $\mathcal{L}_F$ be $\{\lambda_0, \lambda_1, \dots\}$ with corresponding eigenfunctions $\{\psi_0, \psi_1, \dots\}$ and let the eigenvalues of the Schrödinger operator $\mathcal{L}_S$ be $\{\lambda_0, \lambda_1, \dots\}$ (note that we will shortly show that the eigenvalues for Schrödinger operator are equivalent to the eigenvalues for the FPO following, hence the same use of notation) with corresponding eigenfunctions $\{\varphi_0, \varphi_1, \dots\}$. As shown in [20], every FPO $\mathcal{L}_F$ can be transformed into a corresponding Schrödinger operator $\mathcal{L}_S$ with the following transformation,

$$q(x) = e^{U(x)/2D} p(x), \quad (3.13)$$

where $\mathcal{L}_F$ acts on p and $\mathcal{L}_S$ acts on q . We can show this to be true by substituting Eq. (3.13) into Eq. (3.5) to yield

$$\begin{aligned} \mathcal{L}_F[e^{-U(x)/2D} q(x)] &= e^{-U(x)/2D} \left[\frac{d^2 q}{dx^2} - \frac{U_x(x)^2 q(x)}{4D} + \frac{U_{xx}(x) q(x)}{2} \right], \\ &= e^{-U(x)/2D} \left[\frac{d^2 q}{dx^2} - V(x) q(x) \right], \end{aligned}$$

where $U_x = \frac{\partial U}{\partial x}$, $U_{xx} = \frac{\partial^2 U}{\partial x^2}$ and

$$V(x) = \frac{U_x(x)^2}{4D} - \frac{U_{xx}(x)}{2}. \quad (3.14)$$

That is,

$$\mathcal{L}_F[e^{-U(x)/2D} q(x)] = e^{-U(x)/2D} \mathcal{L}_S[q(x)],$$

or equivalently,

$$e^{U(x)/2D} \mathcal{L}_F[p(x)] = \mathcal{L}_S[e^{U(x)/2D} p(x)].$$

Therefore the Fokker-Planck eigenvalue problem is the equivalent to the Schrödinger eigenvalue problem,

$$\mathcal{L}_F \psi = \lambda \psi \quad \Leftrightarrow \quad \mathcal{L}_S \varphi = \lambda \varphi, \quad (3.15)$$

where

$$\varphi(x) = e^{U(x)/2D} \psi(x), \quad (3.16)$$

and the Fokker-Planck potential function $U(x)$ is related to the Schrödinger potential function $V(x)$ by the transformation given in Eq. (3.14).

The equivalence between the eigenvalue problems of the FPO and the Schrödinger operator conveniently allows results from one system to be applied to the other. Note that this equivalence does not extend beyond eigenvalue problems. The potential function $U(x)$ of the FPO

corresponding to a given Schrödinger equation with potential function $V(x)$ can be found as such [20],

$$U(x) = -2D \ln(\varphi_0(x)) + C, \quad (3.17)$$

or alternatively,

$$U_x(x) = -2D \frac{\varphi'_0(x)}{\varphi_0(x)},$$

where φ_0 is the zero eigenfunction (also called *ground-state* eigenfunction in quantum mechanics) of $\mathcal{L}_S$ with eigenvalue $\lambda_0 = 0$ and C is some constant of integration. Note that C is differentiated out when $U(x)$ is used in the FPO and thus the value of C can be arbitrary.

Eq. (3.16) can be used to easily find the equivalent Schrödinger eigenfunction given knowledge of the Fokker-Planck eigenfunctions and potential function. To find an equation easily expressing the equivalent Fokker-Planck eigenfunction given knowledge of the Schrödinger eigenfunctions we substitute Eq. (3.17) into Eq. (3.16) giving,

$$\psi(x) = e^{-C} \varphi_0(x) \varphi(x), \quad (3.18)$$

where φ_0 is the zero-eigenfunction of the Schrödinger operator with corresponding eigenvalue of $\lambda_0 = 0$. Although it is not guaranteed that a given Schrödinger operator has a zero eigenvalue, we can always vertically shift the potential function to give a zero eigenvalue. This is achieved by subtracting $\lambda^* \varphi$ from each side of the Schrödinger eigenvalue problem, giving $V(x) = V^*(x) + \lambda^*$, where $V^*(x)$ and λ^* are the potential function and lowest eigenvalue, respectively, prior to transformation.

3.5 Common transformations

Transformation to constant diffusion

Up until this point we have assumed that the diffusion coefficient D in Eq. (3.1) is constant over space for a given system. Applying the following transformation, given by Risken and Frank [20], allows any FPE with a spatially dependent diffusion coefficient to be rescaled such that the diffusion coefficient (in the transformed system) is indeed constant. Let us consider a FPE with spatial variable x and spatially dependent diffusion coefficient $D(x)$,

$$\frac{\partial p}{\partial t} = \frac{\partial^2}{\partial x^2} [D(x)p(x, t)] - \frac{\partial}{\partial x} [f(x, \mu)p(x, t)]. \quad (3.19)$$

Consider the following change of variable,

$$y = \int_{x_0}^x \sqrt{\frac{\tilde{D}}{D(s)}} ds.$$

Writing Eq. (3.19) in terms of y gives the transformed FPE,

$$\frac{\partial \tilde{p}}{\partial t} = \tilde{D} \frac{\partial^2 \tilde{p}}{\partial y^2} - \frac{\partial}{\partial y} [\tilde{f}(y, \mu) \tilde{p}(y, t)],$$

where $\tilde{D}$, $\tilde{f}(y, \mu)$ and $\tilde{p}(y, t)$ are the transformed diffusion coefficient, drift function and PDF, respectively. $\tilde{D}$, $\tilde{f}(y, \mu)$ and $\tilde{p}(y, t)$ are related to the origin FPE by the following,

$$\tilde{D} = \left(\frac{dy}{dx} \right)^2 D(x) \quad (= \text{const.}),$$

$$\tilde{f}(y, \mu) = \frac{d^2 y}{dx^2} D(x) + \frac{dy}{dx} f(x, \mu),$$

$$\tilde{p}(y, t) = \sqrt{\frac{D(x)}{\tilde{D}}} p(x, t).$$

Therefore it is reasonable to continue to assume that D is constant throughout the remainder of the thesis.

Moving frame of reference

It may be the case that we wish to apply a moving frame of reference to our system. If the location of the important dynamics (for example, location of the basin(s) of attraction) of our system are moving, we may wish to apply a translation to centre the dynamics of interest about the origin. In the case of R-tipping (in which the rate at which the basin of attraction shifts causes tipping) this is particularly true.

By *Itô's formula* (see [17]), if we apply the change of variables

$$Y(t) = g(X(t), t)$$

to the random process $X(t)$, the SDE governing $Y(t)$ is given by

$$dY(t) = g_t(X(t), t) dt + g_x(X(t), t) dX(t) + \frac{1}{2} g_{xx}(X(t), t) (dX(t))^2, \quad (3.20)$$

where $g_t = \frac{\partial g}{\partial t}$, $g_x = \frac{\partial g}{\partial x}$, and $g_{xx} = \frac{\partial^2 g}{\partial x^2}$.

Consider the case where $g(X(t), t)$ applies a translation to centre the system about the stable equilibrium $x^*(t)$ of f (i.e $f(x^*(t), \mu(t)) = 0$) where $X(t)$ is given by Eq. (3.1). That is,

$$g(X(t), t) = X(t) - x^*(t). \quad (3.21)$$

Immediately we see that $g_t = -\frac{dx^*}{dt}$, $g_x = 1$ and $g_{xx} = 0$. Therefore the SDE governing the translated random process $Y(t) = g(X(t), t)$ is given by Eq. (3.20) as,

$$\begin{aligned} dY(t) &= -\frac{dx^*}{dt}dt + dX(t), \\ &= \left(f(Y(t) + x^*(t), \mu(t)) - \frac{dx^*}{dt} \right) dt + \sigma dW(t). \end{aligned}$$

We also define the new spatial variable y as $y = x - x^*(t)$. Let us define

$$f^*(y, \mu(t)) = f(y + x^*(t), \mu(t)) - \frac{dx^*}{dt} \quad (3.22)$$

as the drift function associated with $Y(t)$, that is we can write $dY(t)$ as such,

$$dY(t) = f^*(Y(t), \mu(t)) dt + \sigma dW(t). \quad (3.23)$$

This appears to be a very similar SDE to our original problem given in Eq. (3.1). However there is a subtle difference, although $x^*(t)$ is a stable equilibrium of f , as $x^*(t)$ is shifting over time we find that the system $X(t)$ lags behind, placing the ‘effective’ stable equilibrium away from $x^*(t)$. This effect is illustrated in Fig. 3.2.

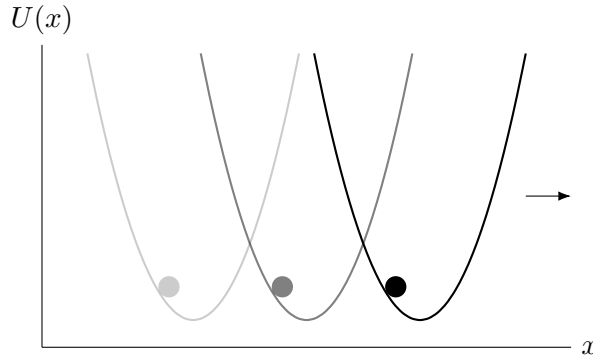


Figure 3.2: An illustrative plot in which the state variable, represented by a circle, lags behind the stable equilibrium as the potential function $U(x)$ is shifted over time to the right.

Mathematically this lag appears in Eq. (3.22) as a $-\frac{dx^*}{dt}$ term. Therefore, although Eq. (3.23) centres $Y(t)$ about the stable equilibrium $x^*(t)$ of the drift function f (i.e. $f(y + x^*(t), \mu(t)) = 0$ at $y = 0$), the $\frac{dx^*}{dt}$ term ‘pushes’ the effective stable equilibrium away from $y = 0$. Let us call the effective stable equilibrium $y^*(t)$, which satisfies $f^*(y^*(t), \mu(t)) = 0$.

Following a similar approach to Berglund and Gentz [5], for general f , x^* , and $\frac{dx^*}{dt}$, with the assumptions that $0 < |\frac{dx^*}{dt}| \ll 1$, i.e. a small rate of change of x^* , we can assume

that the basin of attraction is well approximated by a first order Taylor series about $y = 0$ (or equivalently $x = x^*$) in a manner similar to OU linearisation. Let $f_x = \frac{\partial f}{\partial x}$. Solving $f^*(y^*(t), \mu(t)) = 0$ with first order approximation of f^* yields,

$$y^*(t) \approx \frac{\frac{dx^*}{dt}}{f_x(x^*(t), \mu(t))}.$$

As $Y(t)$ is centred about $x^*(t)$, we can think of $y^*(t)$ as how far the effective stable equilibrium of $Y(t)$ (and thus $X(t)$) is lagging behind $x^*(t)$. Although an approximation was shown here for a general f , it should be noted that for a given f one may be able to solve exactly for $y^*(t)$.

Inverted potential function

We take the following methods for inverted potential function from Risken and Frank [20]. Let us define the *inverted potential function* $\bar{U}$ of U as,

$$\bar{U}(x, \mu) = -U(x, \mu).$$

Let $\bar{\mathcal{L}}_F$ be the FPO with potential function $\bar{U}$, but the same positive drift constant D as $\mathcal{L}_F$. Given that some FPO $\mathcal{L}_F$ has the known eigenvalues $\{\lambda_0, \lambda_1, \dots\}$ and corresponding eigenfunctions $\{\psi_0, \psi_1, \dots\}$, we may be interested in finding the eigenvalues and eigenfunctions of $\bar{\mathcal{L}}_F$. Let $\bar{\mathcal{L}}_F$ have eigenvalues $\{\bar{\lambda}_0, \bar{\lambda}_1, \dots\}$ and corresponding eigenfunctions $\{\bar{\psi}_0, \bar{\psi}_1, \dots\}$.

Inverting the potential function also effects the boundary conditions of the operator, with an absorbing boundary condition in the original problem becoming a reflecting boundary condition in the inverted problem and vice versa. That is, for some FPO $\mathcal{L}_F$ acting on some function $p(x, t)$ with a boundary condition at $x = a$ we find the following relation with the inverted FPO $\bar{\mathcal{L}}_F$ acting on some function $\bar{p}(x, t)$,

$$J(a, t) = 0 \quad \Leftrightarrow \quad \bar{p}(a, t) = 0, \quad (3.24)$$

or alternatively,

$$p(a, t) = 0 \quad \Leftrightarrow \quad \bar{J}(a, t) = 0, \quad (3.25)$$

where J and $\bar{J}$ are the probability fluxes of the original and inverted systems, respectively (see Eq. (3.7)).

We should also remind ourselves of an important property of boundary conditions; two reflecting boundary conditions results in a steady state solution and thus a zero eigenvalue $\lambda_0 = 0$. If either of the boundary conditions are not reflecting then no steady state solution exists and there is no zero eigenvalue.

Let us define the linear operator A acting on some function p as such,

$$Ap = \sqrt{D}e^{-U/2D} \frac{\partial}{\partial x} [e^{U/2D} p].$$

For $n = 0, 1, 2, \dots$ the eigenvalues and eigenfunctions of $\bar{\mathcal{L}}_F$ are related to $\mathcal{L}_F$ by the following cases:

- (i) Given two reflecting boundary conditions (which guarantees that $\lambda_0 = 0$) on the original problem, for the inverted problem we find,

$$\bar{\lambda}_n = \lambda_{n+1} < 0, \quad \bar{\psi}_n = \frac{A\psi_{n+1}}{\sqrt{\lambda_{n+1}}}, \quad (3.26)$$

where the inverted problem has two absorbing boundary conditions.

- (ii) Given two absorbing boundary conditions on the original problem, for the inverted problem we find,

$$\bar{\lambda}_0 = 0, \quad \bar{\lambda}_{n+1} = \lambda_n < 0, \quad \bar{\psi}_{n+1} = \frac{A\psi_n}{\sqrt{\lambda_n}}, \quad (3.27)$$

where the inverted problem has two reflecting boundary conditions.

- (iii) Given one absorbing and one reflecting boundary condition on the original problem, for the inverted problem we find,

$$\bar{\lambda}_n = \lambda_n < 0, \quad \bar{\psi}_n = \frac{A\psi_n}{\sqrt{\lambda_n}}, \quad (3.28)$$

where the inverted problem has one absorbing and one reflecting boundary condition (in the opposite positions to the original problem).

Chapter 4

Eigenmode expansion

We now dedicate a chapter to the framework and details of the *eigenmode expansion* of solutions to the FPE. An *eigenmode* is a spatiotemporal object consisting of an eigenfunction and a corresponding eigenvalue decay term. In this thesis the definition of an eigenmode will cover time-dependent instantaneous eigenfunctions and eigenvalues, as well as the conventional time-independent eigenfunctions and eigenvalues. To speak broadly, the solution to the FPE can be expressed as a linear combination of eigenmodes.

We will divide the time dependence of our setting into two distinct regimes, the *adiabatic regime* and the *non-adiabatic regime*. A time-dependent process in an *adiabatic regime* has gradually changing conditions, allowing the distribution to adapt to the lowest eigenmode [13]. Specifically we consider the limit $\varepsilon \rightarrow 0$ for some rate of change parameter ε . In contrast, a time-dependent process in a *non-adiabatic regime* has rapidly changing conditions, preventing the distribution from adapting to the lowest eigenmode. As a result, the distribution in a non-adiabatic regime will end in a linear combination of eigenmodes.

Sec. 4.1 covers the general machinery of eigenmode expansions. Sec. 4.2 covers the adiabatic regime, while Sec. 4.3 covers the non-adiabatic regime.

4.1 General framework

When studying the FPO at a given time t , and thus a given $\mu(t)$, we will drop the t and $\mu(t)$ dependencies notation for the time being. That is, we let $U(x, \mu(t)) = U(x)$ and $p(x, t) = p(x)$. An initial point of interest is the study of invariant or steady-state-type solutions of $\mathcal{L}_F$ for a given diffusion constant and time t . Let us define the function $\pi(x)$ as,

$$\pi(x) = e^{-U(x)/D}. \quad (4.1)$$

Applying $\mathcal{L}_F$ to $\pi(x)$ yields,

$$\begin{aligned}\mathcal{L}_F \pi &= D \frac{\partial^2}{\partial x^2} e^{-U(x)/D} + \frac{\partial}{\partial x} \left[\frac{\partial U}{\partial x} e^{-U(x)/D} \right], \\ &= \frac{\partial}{\partial x} \left[-\frac{\partial U}{\partial x} e^{-U(x)/D} + \frac{\partial U}{\partial x} e^{-U(x)/D} \right], \\ &= 0.\end{aligned}$$

Therefore $\pi(x)$ is an invariant solution to the FPE for a given t . It should be noted that although π does indeed satisfy $\mathcal{L}_F \pi = 0$, this may or may not be a valid solution to a given system, depending on the boundary conditions. In the case where $\pi(x)$ does agree with the boundary conditions of our system in a given domain, then $\pi(x)$ is trivially a zero-eigenfunction of $\mathcal{L}_F$ with the corresponding eigenvalue $\lambda = 0$.

Finite domain

The function $\pi(x)$ will turn out to be a very useful, being reused several times in the following sections. An immediate use appears when rearranging the eigenvalue problem for the FPO $\mathcal{L}_F$ into Sturm-Liouville form for a given t . For some instantaneous eigenvalue λ and instantaneous eigenfunction ψ , applying the integrating factor $1/\pi(x)$ to $\mathcal{L}_F \psi = \lambda \psi$ yields the following,

$$\frac{\partial}{\partial x} \left[\frac{1}{\pi(x)} D \frac{d\psi}{dx} \right] + \frac{1}{\pi(x)} U(x, t) \psi = \frac{1}{\pi(x)} \lambda \psi. \quad (4.2)$$

We call Eq. (4.2) the *Sturm-Liouville form* of the FPO eigenvalue problem. By classical Sturm-Liouville results presented in Sec. 3.3, given the following conditions are satisfied by Eq. (4.2) on the finite domain $x \in [a, b]$, we have a *regular* Sturm-Liouville problem,

- (i) The functions U , $1/\pi(x)$ and $\frac{d}{dx} 1/\pi(x)$ are all continuous on $[a, b]$.
- (ii) $1/\pi(x) > 0$ for all $x \in [a, b]$.
- (iii) The problem has boundary conditions of the form,

$$\begin{cases} \alpha_1 \psi(a) + \alpha_2 \psi'(a) = 0, & \alpha_1, \alpha_2 \text{ not both } 0 \\ \beta_1 \psi(b) + \beta_2 \psi'(b) = 0, & \beta_1, \beta_2 \text{ not both } 0 \end{cases}$$

Several known properties are then known about $\mathcal{L}_F$, ψ and λ by the Sturm-Liouville Theorem (see Sec. 3.3), notably:

- (i) The eigenvalues $\lambda_0, \lambda_1, \dots$ are real, non-positive, distinct, and can be numbered such that,

$$0 \geq \lambda_0 > \lambda_1 > \dots \rightarrow -\infty$$

- (ii) The normalised eigenfunctions $\{\psi_0, \psi_1, \dots\}$ form an orthonormal basis of the $1/\pi(x)$ weighted Hilbert space $\mathcal{H} = L^2([a, b], 1/\pi(x)dx)$, that is,

$$\langle \psi_n, \psi_m \rangle = \int_a^b \frac{\psi_n(x)\psi_m(x)}{\pi(x)} dx = \delta_{nm}$$

where δ_{nm} is the Kronecker delta. Therefore, any square integrable function $p \in \mathcal{H}$ can be expressed as a linear combination of the eigenfunctions of $\mathcal{L}_F$ as such,

$$p = \sum_{i=0}^{\infty} a_i \psi_i$$

Infinite domain

It should be noted that a Sturm-Liouville problem defined on the infinite domain $x \in (-\infty, \infty)$ is not a regular Sturm-Liouville problem. None the less, there are cases where we wish to study the FPO on an infinite domain, thus it is useful to extend some of the properties of regular Sturm-Liouville problems to the infinite domain. We begin with self-adjointness.

Consider the weighted L^2 Hilbert space $\mathcal{H} = L^2(\mathbb{R}, 1/\pi(x)dx)$ (i.e. infinite domain). The weighted inner product $\langle \cdot, \cdot \rangle$ is defined as follows,

$$\langle p, q \rangle = \int_{-\infty}^{\infty} \frac{p(x)q(x)}{\pi(x)} dx, \quad p, q \in \mathcal{H}. \quad (4.3)$$

We now show that the FPO $\mathcal{L}_F : \mathcal{H} \rightarrow \mathcal{H}$ is self-adjoint with respect to $\langle \cdot, \cdot \rangle$ given certain boundary conditions. Let $p, q \in \mathcal{H}$. We apply integration by parts twice to $\langle \mathcal{L}_F p, q \rangle$.

$$\begin{aligned} \langle \mathcal{L}_F p, q \rangle &= \int_{-\infty}^{\infty} \left(D \frac{\partial^2 p}{\partial x^2} + \frac{\partial}{\partial x} \left[\frac{\partial U}{\partial x} p \right] \right) q e^{U(x)/D} dx, \\ &= \left[\left(D \frac{\partial p}{\partial x} + \frac{\partial U}{\partial x} p \right) q e^{U(x)/D} \right]_{-\infty}^{\infty} - \left[\left(D \frac{\partial q}{\partial x} + \frac{\partial U}{\partial x} q \right) p e^{U(x)/D} \right]_{-\infty}^{\infty} \\ &\quad + \int_{-\infty}^{\infty} p \left(D \frac{\partial^2 q}{\partial x^2} + \frac{\partial}{\partial x} \left[\frac{\partial U}{\partial x} q \right] \right) e^{U(x)/D} dx, \\ &= \left[\left(D \frac{\partial p}{\partial x} + \frac{\partial U}{\partial x} p \right) q e^{U(x)/D} \right]_{-\infty}^{\infty} - \left[\left(D \frac{\partial q}{\partial x} + \frac{\partial U}{\partial x} q \right) p e^{U(x)/D} \right]_{-\infty}^{\infty} + \langle p, \mathcal{L}_F q \rangle. \end{aligned}$$

Therefore given the boundary conditions

$$\frac{\left(D \frac{\partial p}{\partial x} + \frac{\partial U}{\partial x} p(x) \right) q(x)}{\pi(x)} \rightarrow 0 \quad \text{as } |x| \rightarrow \infty \quad \text{for all } p, q \in \mathcal{H}, \quad (4.4)$$

we arrive at the self-adjoint property,

$$\langle \mathcal{L}_F p, q \rangle = \langle p, \mathcal{L}_F q \rangle.$$

The boundary conditions presented in Eq. (4.4) may initially seem rather obscure, but they have a sensible meaning after closer inspection. Recall from Eq. (3.7) that the probability flux $J(x)$ for a distribution $p(x)$ at given time t is given by $J = -D \frac{\partial p}{\partial x} - p \frac{\partial U}{\partial x}$. Therefore, the boundary conditions Eq. (4.4) can be restated as,

$$\frac{J(x)q(x)}{\pi(x)} \rightarrow 0 \quad \text{as } |x| \rightarrow \infty \quad \text{for all } p, q \in \mathcal{H},$$

where $J(x)$ is the probability flux of $p(x)$. Thus Eq. (4.4) simply requires that the product of the probability flux and the probability density of any two functions decays to zero at least as quickly as the invariant function $\pi(x)$.

The self-adjoint property of $\mathcal{L}_F$ (along with sufficient smoothness and positivity of $\pi(x)$) revives some properties already mentioned for regular Sturm-Liouville operators, however with extended domain of $\mathbb{R}$, rather than a finite interval in the case of regular Sturm-Liouville operators (see Sec. 3.2). Self-adjointness implies that the eigenvalues of $\mathcal{L}_F$ are real. As stated in Risken-Frank [20], the eigenvalues of $\mathcal{L}_F$ are also non-positive. Furthermore, the self-adjointness guarantees that eigenfunctions corresponding to distinct eigenvalues are orthogonal with respect to the weighted inner-product given by Eq. (4.3).

As stated by Bennewitz [4], on the infinite domain self-adjointness of an unbounded operator is not a sufficient condition to produce an orthonormal basis of eigenfunctions (despite each eigenfunction being orthogonal). However, we can utilise existing results relating to the Schrödinger operator and their relation to the FPO (as shown in Sec. 3.4) to produce a (less general) sufficient condition for which the FPO has an orthonormal basis of eigenfunctions. As stated by Simon [25], if a Schrödinger operator $\mathcal{L}_S$ (as defined by Eq. (3.12)) has a potential function $V(x)$ satisfying

$$\lim_{|x| \rightarrow \infty} V(x) = \infty, \tag{4.5}$$

then $\mathcal{L}_S$ has orthonormal eigenfunctions $\{\varphi_0, \varphi_1, \dots\}$ forming an basis of (unweighted) square integrable functions. That is, any function $q(x)$ satisfying

$$\int_{-\infty}^{\infty} q(x)^2 dx < \infty,$$

can be expressed in terms of the eigenfunctions of $\mathcal{L}_S$, given that $\mathcal{L}_S$ has a potential function satisfying Eq. (4.5). Therefore, by Eqs. (3.14) and (3.15) we can say that if a FPO $\mathcal{L}_F$ has a potential function $U(x)$ satisfying

$$\lim_{|x| \rightarrow \infty} \frac{U_x(x)^2}{4D} - \frac{U_{xx}(x)}{2} = \infty, \tag{4.6}$$

where $U_x = \frac{\partial U}{\partial x}$ and $U_{xx} = \frac{\partial^2 U}{\partial x^2}$, then $\mathcal{L}_F$ has orthonormal eigenfunctions $\{\psi_0, \psi_1, \dots\}$ forming an basis of (weighted) square integrable functions. The weight function appears when applying the transformation Eq. (3.16) relating the corresponding eigenfunctions of the Schrödinger operator and the FPO. The weight function happens to be $1/\pi(x)$, the same weight function yielding the self adjoint inner product given by Eq. (4.3). Thus, the normalised eigenfunctions $\{\psi_0, \psi_1, \dots\}$ of $\mathcal{L}_F$ form an orthonormal basis of the $1/\pi(x)$ weighted Hilbert space $\mathcal{H} = L^2(\mathbb{R}, 1/\pi(x)dx)$ given that $\mathcal{L}_F$ has a potential function satisfying Eq. (4.6). That is, any function $p \in \mathcal{H}$ can be expressed as a linear combination of eigenfunctions,

$$p = \sum_{i=0}^{\infty} a_i \psi_i, \quad (4.7)$$

where each constant a_i is given by projecting p onto ψ_i as such,

$$a_i = \langle \psi_i, p \rangle. \quad (4.8)$$

Polynomial potential functions are popular models which frequently appear in literature. We can quickly show that any polynomial of degree $n \geq 2$ provides a potential function which satisfies Eq. (4.6). Let $U(x) = P_n(x)$ be a potential function given by some polynomial of degree $n \geq 2$. When computing the limit given by Eq. (4.6) it is sufficient to concern ourselves with only the sign and degree of the highest degree term of $U_x(x)^2/4D - U_{xx}(x)/2$ (as the highest degree term will dominate in the limit). Given that $U(x) = P_n(x)$, we can conclude the following,

$$\deg(U_x(x)^2) = 2(n-1), \quad \deg(U_{xx}(x)) = n-2.$$

Therefore given $n \geq 2$, $\deg(U_x(x)^2) > \deg(U_{xx}(x))$, and the $U_x(x)^2$ term will always dominate in the limit. It is also clear that $\deg(U_x(x)^2)$ is an even degree. We know that $1/4D > 0$ by definition, and we also know that the coefficient of the highest degree term of $U_x(x)^2$ must be positive (as it was squared). Therefore the limit of the largest degree term of $U_x(x)^2/4D$ as $|x| \rightarrow \pm\infty$ is $+\infty$, thus satisfying Eq. (4.6) for polynomial potential functions of degree two or more.

Note that the proof above also extends to potential functions given by a polynomial of degree two or more which are perturbed by another function given certain limit conditions are satisfied. Consider the potential function U given by $U(x) = P_n(x) + V(x)$, where $V(x)$ satisfies the following,

$$\lim_{|x| \rightarrow \infty} V_x(x)^2 < \infty, \quad \lim_{|x| \rightarrow \infty} V_{xx}(x) < \infty.$$

Under such conditions $U(x)$ still satisfies Eq. (4.6). An example of such a function $V(x)$ is a Gaussian type function $V(x) = e^{-x^2}$.

Eigenmode formulation

We have established that on the finite domain $[a, b]$, given Sturm Liouville conditions are satisfied, or on the infinite domain $\mathbb{R}$, given Eq. (4.6) is satisfied, the eigenfunctions of $\mathcal{L}_F$ form an orthonormal basis of the given Hilbert space $\mathcal{H}$. It should be noted that in the case of a time-dependent FPO $\mathcal{L}_F$ that its instantaneous eigenfunctions and eigenvalues will also evolve over time. Therefore if we wish to represent a given probability density function p in terms of the eigenfunctions of $\mathcal{L}_F$, then the basis coefficients in this representation will be affected over time by both the changes in p and the changes in the basis of eigenfunctions. That is, a given eigenvalue is time dependent, i.e. $\lambda_n(t)$, and a given eigenfunction is space and time dependent, i.e. $\psi_n(x, t)$. Let us take a closer look at dynamics of this time evolution of basis coefficients.

Consider the representation of some probability density function $p(x, t)$ in terms of instantaneous eigenfunctions $\{\psi_0, \psi_1, \dots\}$ of $\mathcal{L}_F$,

$$p(x, t) = \sum_{i=0}^{\infty} a_i(t) \psi_i(x, t). \quad (4.9)$$

As presented by Ritchie and Sieber [21, 22], we can compute the time evolution of each coefficient $a_n(t)$. By definition,

$$\frac{\partial p}{\partial t} = \mathcal{L}_F p.$$

Substituting Eq. (4.9) we obtain,

$$\frac{\partial}{\partial t} \left[\sum_{i=0}^{\infty} a_i \psi_i \right] = \mathcal{L}_F \left[\sum_{i=0}^{\infty} a_i \psi_i \right].$$

Differentiating and applying the properties of eigenfunctions yields,

$$\sum_{i=0}^{\infty} \frac{da_i}{dt} \psi_i + \sum_{i=0}^{\infty} a_i \frac{\partial \psi_i}{\partial t} = \sum_{i=0}^{\infty} a_i \lambda_i \psi_i.$$

Apply $\langle \psi_n, \cdot \rangle$ to each side and cancel terms by using the orthogonality of eigenfunctions with respect to $\langle \cdot, \cdot \rangle$,

$$\frac{da_n}{dt} = a_n(t) \lambda_n(t) - \sum_{i=0}^{\infty} a_i(t) \langle \psi_n(x, t), \frac{\partial \psi_i}{\partial t} \rangle. \quad (4.10)$$

We refer to Eq. (4.9) as the *eigenmode expansion* of a given solution p to the FPE. Eqs. (4.9) and (4.10) form an important tool for the study of solutions to the FPE. For some initial condi-

tion $p(x, 0) = \delta(x - x_0)$ we can compute the initial condition $a_n(0)$ using Eq. (4.8) as such,

$$\begin{aligned} a_n(0) &= \int_a^b \frac{\delta(x - x_0) \psi_n(x, 0)}{\pi(x)} dx \\ &= \frac{\psi_n(x_0, 0)}{\pi(x_0)}. \end{aligned} \quad (4.11)$$

In the case that the eigenmode expansion for a given solution p is truncated, giving an approximated solution, we define the n -mode expansion p_n as follows,

$$p_n(x, t) = \sum_{i=0}^{n-1} a_i(t) \psi_i(x, t) \quad (4.12)$$

The following two sections are motivated by the effect of time scales in different regimes on Eq. (4.10).

4.2 Adiabatic regime

We now focus specifically on the case of the adiabatic regime. Informally, we can think of the adiabatic regime (in the context of the FPE) as the case in which the temporal rate of change in potential function U is so small that U can be assumed to be constant over an interval ‘near’ some time t . Removing the time dependence from U allows simplifications to be made to the eigenmode expansion of solutions to the FPE. Tipping point systems are frequently studied through an adiabatic lens, for example, Kuehn’s fast-slow framework [16]. We now recall and apply the linear ramping of the bifurcation parameter $\mu(t)$ from Kuehn’s fast-slow framework,

$$d\mu = \varepsilon dt, \quad (4.13)$$

where ε is the *rate of change parameter*. In this thesis, we consider an adiabatic regime to be a Fokker-Planck system in the limit $\varepsilon \rightarrow 0$. We introduce a second time scale τ to aid analysis. Let us call t the *fast time scale* and τ the *slow time scale*. The probability density function of the system evolves at the fast time scale while the potential function evolves at the slow time scale. We define τ as such,

$$\tau = \varepsilon t.$$

Therefore,

$$\frac{\partial U}{\partial t} = \varepsilon \frac{\partial U}{\partial \tau}.$$

Thus in the limit $\varepsilon \rightarrow 0$ we see

$$\frac{\partial U}{\partial t} \rightarrow 0.$$

Behaviour in fast time scale

Therefore, in the adiabatic regime the potential function U is approximately constant with respect to the fast time scale t . For a given τ we let $U(x, t) = U(x)$ and we can assume time independence in the fast time scale. It immediately follows that $\mathcal{L}_F$, its eigenfunctions, and eigenvalues are also time independent with respect to the fast time scale. Consequently Eq. (4.10) can be simplified as such,

$$\frac{da_n}{dt} = \lambda_n a_n. \quad (4.14)$$

Unlike the full form of Eq. (4.10), following the adiabatic simplification, Eq. (4.14) can be solved in closed form for solution, giving,

$$a_n(t) = a_n(0)e^{\lambda_n t},$$

where $a_n(0)$ is the initial condition for a_n . In the case of a delta function initial condition, $a_n(0)$ is given by Eq. (4.11). The full probability density function p can be written as follows,

$$p(x, t) = \sum_{i=0}^{\infty} a_n(0)e^{\lambda_i t} \psi_n(x). \quad (4.15)$$

This is a common solution found when studying linear partial differential equations that allow a separation of variables. Notice that over time, the eigenmodes with negative corresponding eigenvalues will decay away. Furthermore, eigenmodes with a corresponding zero eigenvalue will cause the eigenmode to persist without growth or decay. We recall that the eigenvalues of the FPO are non-positive, thus eliminating the possibility of growth.

Behaviour in slow time scale

For any significant period of time we can see that only the eigenmode corresponding to the zero eigenvalue persists (if it exists). That is to say, p follows the zero-eigenmode (the steady state solution) for all $\tau > 0$. All higher eigenmodes decay and have no contribution.

$$p(x, \tau/\varepsilon) = a_0(\tau/\varepsilon) \psi_0(x, \tau/\varepsilon). \quad (4.16)$$

This is an intuitive result, we often think of systems in slow time scales being well described by a steady state solution. We must note that $\psi_0(x)$ is normalised with respect to the weighted norm $\|\psi_0\| = \sqrt{\langle \psi_0, \psi_0 \rangle} = 1$ given by Eq. (4.3). As p is a steady state PDF we also require that $\int_a^b p dx = 1$, therefore $a_0(\tau/\varepsilon)$ is given by,

$$a_0(\tau/\varepsilon) = \left(\int_a^b \psi_0(x, \tau/\varepsilon) dx \right)^{-1}.$$

Tipping rates in fast time scale

Given a system with a bistable potential function has relaxed to the lowest eigenmode (that is, it is in a steady state), we can employ the lowest non-zero eigenvalue to approximate transition rates between the two basins of attraction. This can be thought of as the rate at which the system is tipping (in a particular direction). For a system with a clearly defined bistable potential function, we can label the probability density in the left well as p_L and the right well as p_R . All probability density is labelled as a member of either the left well or the right well. That is,

$$p_L + p_R = 1. \quad (4.17)$$

Let the rate at which particles transition from the left well to the right be r_{LR} , and the rate from right to left be r_{RL} . By detailed balance,

$$p_L r_{LR} = p_R r_{RL} \quad (4.18)$$

By [20], in a deep, well separated bistable potential function,

$$\lambda_1 \approx r_{LR} + r_{RL}. \quad (4.19)$$

Combining Eqs. (4.17) to (4.19) gives,

$$r_{RL} \approx p_L \lambda_1,$$

and,

$$r_{LR} \approx p_R \lambda_1.$$

In the case of an even potential function, $p_R = p_L = \frac{1}{2}$ and thus $r = r_{RL} = r_{LR}$ is given by,

$$r \approx \frac{\lambda_1}{2}. \quad (4.20)$$

4.3 Non-adiabatic regime

In real world systems, a completely adiabatic regime is never achieved (unless the FPO governing the system does not vary with time, in which case no tipping phenomena of the sort we are interested in is observed). As a result, we often find ourselves applying the properties of an adiabatic system (presented in Sec. 4.2) to a non-adiabatic system. That is to say, in the system where the potential function U is in fact (slowly) changing, the assumption that U is constant over time produces errors. In addition, the full eigenmode expansion presented in Eq. (4.9) must be truncated at some finite number of modes (see Eq. (4.12)) in order to compute explicit solutions, also producing errors. In this section, we consider such errors that arise from both adiabatic assumptions and from truncations of the eigenmode expansion.

For a given time t , consider the n -mode truncated approximation given by Eq. (4.12),

$$p_n(x) = \sum_{i=0}^{n-1} a_i \psi_i(x)$$

where a_i is a scalar. The L_2 error between p_n and the true PDF p is given by,

$$\begin{aligned} \|p - p_n\|_2 &= \sqrt{\int_a^b \left(\sum_{i=0}^{\infty} a_i \psi_i(x) - \sum_{i=0}^{n-1} a_i \psi_i(x) \right)^2 dx} \\ &= \sqrt{\int_a^b \sum_{i=n}^{\infty} a_i^2 \psi_i^2(x) + \sum_{i=n}^{\infty} \sum_{j>i} 2a_i a_j \psi_i(x) \psi_j(x) dx} \\ &= \sqrt{\sum_{i=n}^{\infty} a_i^2 \int_a^b \psi_i^2(x) dx + \sum_{i=n}^{\infty} \sum_{j>i} 2a_i a_j \int_a^b \psi_i(x) \psi_j(x) dx} \\ &= \sqrt{\sum_{i=n}^{\infty} a_i^2} \end{aligned}$$

That is to say, the truncation error (assuming that the basis coefficients which are included are exactly correct) is equal to the L_2 norm of the vector of the truncated basis coefficients.

Chapter 5

Eigenvalues and eigenfunctions of the Fokker-Planck operator

Up to this point, the eigenfunctions and eigenvalues of the FPO have been used extensively without any mention of their actual solutions. The following chapter is dedicated to the solutions of Fokker-Planck eigenvalue problems, both exactly solvable cases and approximated solutions. It is unfortunately the case that only a select group of potential functions yield analytic solutions to their eigenvalue problem, with other cases requiring an approximation of some form. For the entirety of this chapter we assume that we are solving the eigenvalue problem of the FPO for a given μ , and thus drop the μ from $U(x, \mu)$ for the time being.

In Sec. 5.1 we summarise known results to find analytic solutions. In Sec. 5.2 we explore and extend the BCPM, while in Sec. 5.3 the classical Rayleigh-Schrödinger perturbation method is presented.

5.1 Eigenvalues and eigenfunctions of solvable potential functions

There are several potential functions U for which all eigenvalues and eigenfunctions of FPO are known in analytic form. We call these *exactly solvable* potential functions. For all the following examples, eigenvalues and eigenfunctions are indexed by $n = 0, 1, 2, \dots$. Known results are primarily taken from Risken and Frank [20].

Parabolic potential

The *parabolic potential* is given by

$$U(x) = \frac{\alpha}{2}x^2, \tag{5.1}$$

for the parameter $\alpha > 0$. The previously mentioned Ornstein-Uhlenbeck process has a probability density function governed by the FPE with a parabolic potential. Similar to many of the following exactly solvable potential functions, the parabolic potential finds its solution from a transformation of the Schrödinger equation. The Schrödinger equation with a parabolic potential (called the *quantum harmonic oscillator* in quantum mechanics) has a known set of eigenvalues and eigenfunctions, written in terms of Hermite polynomials. Interestingly, the corresponding transformed Fokker-Planck potential function is also parabolic. It is not generally the case that a Schrödinger potential function and the corresponding transformed Fokker-Planck potential function have the same form, making the parabolic potential a special case.

Given the importance of the parabolic potential, we show a derivation of the eigenvalues and eigenfunctions. We begin by introducing the *Hermite polynomials*, a set of orthogonal polynomials. We call $H_n(x)$ the order n Hermite polynomial. It should be noted that any reference to Hermite polynomials in this thesis refers to the *physicist's* Hermite polynomials, rather than the *probabilist's* Hermite polynomials, denoted $He_n(x)$. The Hermite polynomials can be defined by the following recurrence relation,

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x), \quad (5.2)$$

where $H_0(x) = 1$. As such, the first six Hermite polynomials are,

$$\begin{aligned} H_0(x) &= 1, \\ H_1(x) &= 2x, \\ H_2(x) &= 4x^2 - 2, \\ H_3(x) &= 8x^3 - 12x, \\ H_4(x) &= 16x^4 - 48x^2 + 12, \\ H_5(x) &= 32x^5 - 160x^3 + 120x. \end{aligned}$$

There exist many useful properties of Hermite polynomials, a few of which will now be stated without proof:

- (i) Hermite polynomials are orthogonal with respect to the weight function $w(x) = e^{-x^2}$. That is,

$$\int_{-\infty}^{\infty} H_n(x)H_m(x)w(x)dx = \sqrt{\pi} 2^n n! \delta_{nm}, \quad (5.3)$$

where δ_{nm} is the Kronecker delta.

- (ii) The set of Hermite polynomials for an orthogonal basis of the real functions f satisfying the following,

$$\int_{-\infty}^{\infty} f(x)^2 w(x) dx < \infty,$$

with the following inner product,

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(x)g(x)w(x) dx.$$

(iii) The derivative of $H_n(x)$ satisfies,

$$\frac{d}{dx}H_n(x) = 2nH_{n-1}(x),$$

or more generally,

$$\frac{d^m}{dx^m}H_n(x) = \frac{2^m n!}{(n-m)!}H_{n-m}(x).$$

The *Hermite functions* (or *Hermite-Gaussian functions*), denoted $h_n(x)$, are formed by transforming the Hermite polynomials as such,

$$h_n(x) = e^{-x^2/2}(2^n n! \sqrt{\pi})^{-1/2} H_n(x). \quad (5.4)$$

Observe that the Hermite functions contain the square root of the weight function $w(x)$ and normalisation term $\sqrt{\pi}2^n n!$ found in Property (i). Therefore, substituting Eq. (5.4) into Eq. (5.3) gives the following,

$$\int_{-\infty}^{\infty} h_n(x)h_m(x)dx = \delta_{nm}. \quad (5.5)$$

That is, the Hermite functions form an orthonormal basis under the unweighted inner product. A notable property of the Hermite functions is that they satisfy the following differential equation,

$$\frac{d^2}{dy^2}h_n(y) - (y^2 - 1)h_n(y) = -2nh_n(y). \quad (5.6)$$

We are starting to see an equation reminiscent of the Schrödinger operator eigenvalue problem. Let us continue in this vein and perform the following rescaling of y ,

$$y = \sqrt{\frac{\alpha}{2D}}x, \quad (5.7)$$

and thus,

$$\frac{d^2}{dy^2} = \frac{2D}{\alpha} \frac{d^2}{dx^2}. \quad (5.8)$$

We must also scale $h_n(y)$ by $\left(\frac{\alpha}{2D}\right)^{\frac{1}{4}}$ to preserve the normalisation property of Eq. (5.5) following the change of variables Eq. (5.7). Substituting Eqs. (5.7) and (5.8) into Eq. (5.6) and rearranging gives,

$$D \frac{d^2}{dx^2} \left(\frac{\alpha}{2D}\right)^{\frac{1}{4}} h_n(y) - \alpha \left(\frac{\alpha x^2}{4D} - \frac{1}{2}\right) \left(\frac{\alpha}{2D}\right)^{\frac{1}{4}} h_n(y) = -\alpha n \left(\frac{\alpha}{2D}\right)^{\frac{1}{4}} h_n(y),$$

or equivalently,

$$D \frac{d^2}{dx^2} \varphi_n(x) - V(x) \varphi_n(x) = \lambda_n \varphi_n(x), \quad (5.9)$$

where $\lambda_n = -\alpha n$, $\varphi_n(x) = \left(\frac{\alpha}{2D}\right)^{\frac{1}{4}} h_n(y)$, $y = \sqrt{\frac{\alpha}{2D}} x$, and $V(x) = \alpha \left(\frac{\alpha x^2}{4D} - \frac{1}{2}\right)$. Notice that Eq. (5.9) is exactly the Schrödinger eigenvalue problem. That is, $\lambda_n = -\alpha n$ is an eigenvalue with corresponding normalised eigenfunction $\varphi_n(x) = \left(\frac{\alpha}{2D}\right)^{\frac{1}{4}} h_n(y)$ of the Schrödinger operator with potential function $V(x) = \alpha \left(\frac{\alpha x^2}{4D} - \frac{1}{2}\right)$. This potential function is of no accident. Let us recall the parabolic potential $U(x)$ Eq. (5.1) of the FPE which we are concerned about and find the corresponding Schrödinger potential function $V(x)$. By Eq. (3.14) we find,

$$\begin{aligned} V(x) &= \frac{U_x(x)^2}{4D} - \frac{U_{xx}(x)}{2}, \\ &= \alpha \left(\alpha \left(\frac{x}{2D} \right)^2 - \frac{1}{2} \right). \end{aligned}$$

By Eq. (3.15), the Schrödinger eigenvalue problem Eq. (5.9) is the equivalent to the eigenvalue problem for the FPO with parabolic potential given by Eq. (5.1). Thus we can apply the transformation Eq. (3.16) to find the corresponding eigenfunctions for the Fokker-Planck eigenvalue problem.

$$\begin{aligned} \psi_n(x) &= e^{-U(x)/2D} \varphi_n(x), \\ &= e^{-y^2/2} \left(\frac{\alpha}{2D} \right)^{\frac{1}{4}} h_n(y), \\ &= e^{-y^2} \left(\frac{\alpha}{2D} \right)^{\frac{1}{4}} (2^n n! \sqrt{\pi})^{-1/2} H_n(y). \end{aligned}$$

To summarise, the FPO with parabolic potential given by Eq. (5.1) has eigenvalues and (normalised) eigenfunctions (written in terms of the zero-eigenfunction for simplicity) given by,

$$\begin{aligned} \lambda_n &= -\alpha n, \\ \psi_0(x) &= \left(\frac{\alpha}{2\pi D} \right)^{\frac{1}{4}} e^{-y^2}, \\ \psi_n(x) &= \frac{\psi_0(x) H_n(y)}{\sqrt{2^n n!}}, \end{aligned}$$

where $y = \sqrt{\frac{\alpha}{2D}} x$ and H_n is the n th Hermite polynomial. See Fig. 5.1a for an example.

Inverted parabolic potential

Given that we know the eigenvalues and eigenfunctions of the parabolic potential, we can now apply the inverted potential transformation presented in Sec. 3.5. Let us define the inverted parabolic potential as,

$$\bar{U}(x) = \frac{-\alpha}{2}x^2.$$

The resulting eigenvalues and corresponding eigenfunctions are,

$$\begin{aligned}\lambda_n &= -\alpha(n+1), \\ \bar{\psi}_0(x) &= \left(\frac{\alpha}{2\pi D}\right)^{\frac{1}{4}} e^{-y^2}, \\ \bar{\psi}_n(x) &= \frac{\bar{\psi}_0(x)H_n(y)}{\sqrt{2^n n!}},\end{aligned}$$

where $y = \sqrt{\frac{\alpha}{2D}}x$ and H_n is the n th Hermite polynomial. Apart from a α shift in the eigenvalues, the eigenvalues and eigenfunctions of the inverted parabolic potential are identical to the parabolic potential. This is a special case of inverted potentials.

Transformed infinite square well

The infinite square well potential for the Schrödinger equation has the potential function $V(x)$ given by

$$V(x) = \begin{cases} \infty, & x \leq a \\ -\frac{D\pi}{b-a}, & a \leq x \leq b \\ \infty, & x \geq b. \end{cases}$$

Inside the infinite square well (i.e. $x \in (a, b)$) the system behaves like the free particle. The appropriate boundary conditions in the quantum setting are Dirichlet boundary conditions. We find the eigenfunctions of the Schrödinger equation are

$$\varphi_n(x) = \sin\left(\frac{(n+1)\pi(x-a)}{b-a}\right),$$

with eigenvalues

$$\lambda_n = -D \left(\frac{n\pi}{b-a}\right)^2. \quad (5.10)$$

By Eq. (3.14) we find that the corresponding potential function for the FPE is

$$U(x) = -2D \ln\left(\sin\left(\frac{\pi(x-a)}{b-a}\right)\right)$$

on the domain $x \in [a, b]$. By Eq. (3.18), the resulting eigenfunctions for the FPE are

$$\begin{aligned}\psi_n(x) &= \varphi_0(x)\varphi_n(x), \\ &= \sin\left(\frac{\pi(x-a)}{b-a}\right) \sin\left(\frac{(n+1)\pi(x-a)}{b-a}\right),\end{aligned}$$

with the same eigenvalues given by Eq. (5.10). See Fig. 5.1b for an example.

Transformed Pöschl-Teller potential

Another example of a Schrödinger potential with a known set of eigenvalues and eigenfunctions is the Pöschl-Teller potential, give by the potential function

$$V(x) = \frac{1}{2} - \frac{N(N+1)}{2} \operatorname{sech}^2(x),$$

with eigenvalues

$$\lambda_n = \frac{1 - (n+1)^2}{2},$$

and eigenfunctions

$$\varphi_n(x) = P_N^{n+1}(\tanh(x)),$$

where P_l^m is the Legendre function of degree l and order m . The transformed Pöschl-Teller potential $U(x)$ corresponding to the FPO is given by Eq. (3.14) as,

$$U(x) = -2D \log(P_N^{n+1}(\tanh(x)))$$

Following the transformation given by Eq. (3.18) we find the corresponding Fokker-Planck eigenfunctions,

$$\begin{aligned}\psi_n(x) &= \varphi_0(x)\varphi_n(x) \\ &= P_N^1(\tanh(x))P_N^{n+1}(\tanh(x)).\end{aligned}$$

It should be noted that the Pöschl-Teller potential has only N eigenfunctions (and N corresponding eigenvalues). The eigenfunctions do not form a complete basis, it can be seen that $U(x)$ does not satisfy Eq. (4.6).

Free particle

The simplest case of a FPO with a known set of eigenvalues and eigenfunctions is that of the *free particle*. The free particle is a purely diffusive process with no forces from a potential function. That is, we can let the potential function be defined as

$$U(x) = 0.$$

The resulting FPE is the *heat equation* in one spatial dimension,

$$\frac{\partial p}{\partial t} = D \frac{\partial^2 p}{\partial x^2}.$$

The eigenvalues and eigenfunctions of the free particle on the infinite domain are degenerate, as for each eigenvalue there corresponds an infinite number of eigenfunctions (by translation). However, on a finite domain $[a, b]$ with regular Sturm-Liouville boundary conditions, the free particle has a known set of eigenvalues and eigenfunctions in terms of trigonometric functions. We consider the cases of homogeneous Dirichlet (absorbing) and Neumann (reflecting) boundary conditions (or possibly one of each). The eigenvalues and eigenfunctions of the free particle for each combination of boundary conditions are presented in Table 5.1. See Fig. 5.1c for an example with reflecting boundary conditions.

Left BC	Right BC	Eigenvalue λ_n	Eigenfunction $\psi_n(x)$
$\psi_n(a) = 0$	$\psi_n(b) = 0$	$-D \left(\frac{(n+1)\pi}{b-a} \right)^2$	$\sin \left(\frac{(n+1)\pi(x-a)}{b-a} \right)$
	$\psi'_n(b) = 0$	$-D \left(\frac{(n+\frac{1}{2})\pi}{b-a} \right)^2$	$\sin \left(\frac{(n+\frac{1}{2})\pi(x-a)}{b-a} \right)$
$\psi'_n(a) = 0$	$\psi_n(b) = 0$	$-D \left(\frac{(n+\frac{1}{2})\pi}{b-a} \right)^2$	$\cos \left(\frac{(n+\frac{1}{2})\pi(x-a)}{b-a} \right)$
	$\psi'_n(b) = 0$	$-D \left(\frac{n\pi}{b-a} \right)^2$	$\cos \left(\frac{n\pi(x-a)}{b-a} \right)$

Table 5.1: The eigenvalues and eigenfunctions of the free particle FPO for a selection of boundary conditions over the domain $x \in [a, b]$. Note that for simplicity, the unnormalised eigenfunctions are presented here.

5.2 Boundary condition perturbation method

In this section we take inspiration and expand upon the *boundary condition (BC) perturbation method* presented by Ritchie and Sieber [21, 22]. The method provides an approximation for the leading non-zero eigenvalue and the corresponding eigenfunction of the FPO for certain potential functions. As mentioned previously (see Eq. (4.1)), an invariant solution to the FPO can

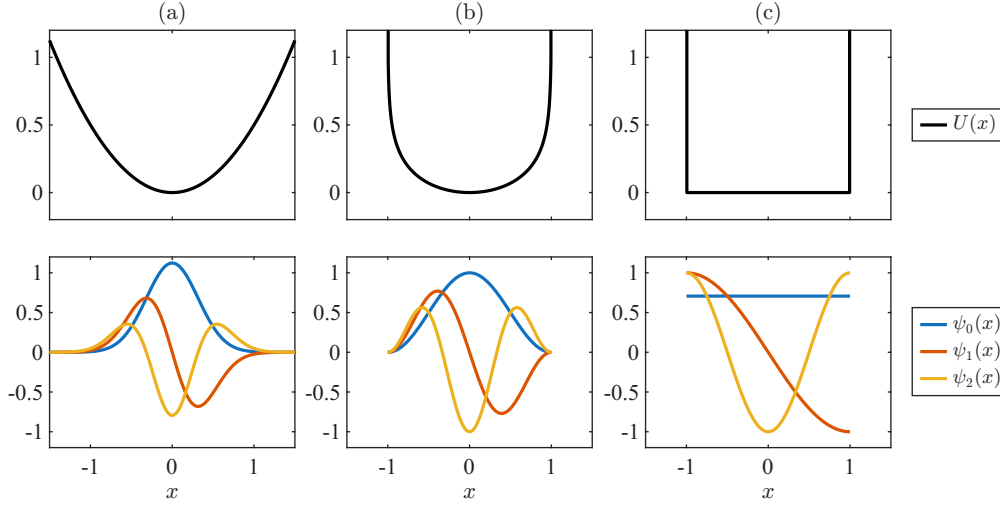


Figure 5.1: Examples of the potential functions (top row) and the corresponding first three eigenfunctions (bottom row) of (a) the parabolic potential, (b) the transformed infinite square well and (c) the free-particle with reflecting boundary conditions.

typically be computed analytically. If this invariant solution satisfies the boundary conditions of the system, then it is an eigenfunction corresponding to the zero eigenvalue. In contrast, apart from the cases mentioned in Sec. 5.1, it is very rare to find analytic solutions to the non-zero eigenvalues and the corresponding eigenfunctions. The BCPM utilises the fact that the leading non-zero eigenvalue is very close to zero for systems with a strong basin of attraction. Following a relaxation of the boundary condition, an analytic solution to the eigenvalue problem can be computed for the zero eigenvalue. The boundary condition is then corrected using a linear perturbation, yielding an approximation of the leading non-zero eigenvalue and the corresponding eigenfunction. The leading non-zero eigenvalue is of particular interest to the study of tipping point systems as it corresponds to tipping rates, for example see Eqs. (4.19) and (4.20).

The BCPM was originally presented by Ritchie and Sieber [21, 22] for asymmetric potential functions U with the property $U(x) \rightarrow \pm\infty$ as $x \rightarrow \infty$ and $U(x) \rightarrow \mp\infty$ as $x \rightarrow -\infty$. That is, a potential function that approaches ∞ at one boundary and approaches $-\infty$ at the opposite boundary. The method then restricts the domain to a sufficiently large domain $[a, b]$. The method also requires a basin of attraction within the potential function from which tipping might occur (see Fig. 5.2a for an example of such a potential function). Ritchie and Sieber originally used a perturbation of the boundary condition itself and then found the relationship between the boundary condition and the eigenvalue to approximate the leading non-zero eigenvalue. In this thesis, we instead directly observe the effect of changes in the eigenvalue (i.e. the

eigenvalue is treated as a perturbation parameter). Although the implementation of the BCPM in this thesis appears different to the original work by Ritchie and Sieber, it follows exactly the same principles and thought process. Furthermore, Ritchie and Sieber applied the BCPM with two absorbing boundary conditions. We also allow for one absorbing boundary condition and one homogeneous Neumann condition. This allows us to extend the method to apply to even potential functions with the form $U(x) \rightarrow \infty$ as $|x| \rightarrow \infty$, and the form $U(x) \rightarrow -\infty$ as $|x| \rightarrow \infty$ (i.e even confining potential functions and even anti-confining potential functions).

Unperturbed solution with free boundary condition

Without loss of generality, let us look at the case of a FPO with an absorbing boundary condition on the left and a undefined boundary condition on the right. We assume that the potential function contains a basin of attraction and that the boundary conditions have been selected in a manner appropriate to the potential function (see Fig. 3.1). In this case the absorbing boundary condition on the left is appropriate if $U(x) \rightarrow -\infty$ as $x \rightarrow -\infty$. The (normalised) eigenvalue problem for the zero eigenvalue for such a system on the domain $x \in [a, b]$ is given as

$$\begin{cases} \mathcal{L}_F \psi = 0, & x \in [a, b], \\ \psi(a) = 0, \\ \int_a^b \psi dx = 1. \end{cases} \quad (5.11)$$

Eq. (5.11) is simpler to solve than a system with two boundary conditions and non-zero eigenvalue (for which an analytical solution is seldom available), and thus allows an analytical solution of ψ to be found. We call this solution $p^{(0)}$. Solving Eq. (5.11) gives,

$$p^{(0)}(x) = c_0 \int_a^x \exp\left(\frac{U(s) - U(x)}{D}\right) ds, \quad (5.12)$$

where the constant c_0 is given by,

$$c_0 = \left(\int_a^b \int_a^{s_2} \exp\left(\frac{U(s_1) - U(s_2)}{D}\right) ds_1 ds_2 \right)^{-1}.$$

We can interpret the solution Eq. (5.12) as a steady state distribution which is achieved by allowing a non-zero probability flux at $x = b$, which then flows towards the absorbing boundary condition at $x = a$. In the following subsections we ‘correct’ the initially undefined boundary condition from solution $p^{(0)}$ to be either an absorbing boundary condition or a homogeneous Neumann boundary condition. In both cases, the initial solution $p^{(0)}$ should act in a ‘similar’ manner at $x = a$ to our final desired boundary condition to avoid errors resulting from large perturbations. As a result, we require the following conditions to be met in order for an accurate approximation,

- (i) There is a relatively large potential well in the domain $x \in [a, b]$ with stable equilibrium at x^* . This criteria ensures that the true eigenvalue is also close to zero, therefore avoiding large errors from our assumption used to compute Eq. (5.12).
- (ii) The probability flux entering at $x = b$ has a larger potential than at $x = a$ and at least as large as $x = x^*$, that is, $U(b) > U(a)$ and $U(b) \geq U(x^*)$. This ensures that a ‘natural’ flow of probability flux occurs from $x = b$ to $x = a$ (very informally, when thinking of a potential function, probability flux naturally flows ‘downhill’), resulting in the initially undefined boundary condition at $x = b$ to be ‘similar’ to an absorbing boundary condition or a homogeneous Neumann boundary condition.

Perturbation of boundary condition

Let us now resolve the undefined right boundary condition provided by the solution $p^{(0)}$ by linear perturbation. By perturbing the $p^{(0)}$ while maintaining the boundary condition of $\psi(a) = 0$ we can find an approximated solution which satisfies two boundary conditions.

The linear perturbed solution $p_{\hat{\lambda}}$ is controlled by the parameter $\hat{\lambda}$, which has a similar role as an eigenvalue λ . $p_{\hat{\lambda}}$ is a perturbation about $\hat{\lambda} = 0$, representing the fact that Eq. (5.11) was solved for $\lambda = 0$ and we wish to perturb away from this solution. The single boundary condition and normalisation condition found in Eq. (5.11) are carried through to $p_{\hat{\lambda}}$. We call the first-order correction in the linear perturbation $p^{(1)}$ and define the perturbed solution $p_{\hat{\lambda}}$ as such,

$$\begin{cases} p_{\hat{\lambda}}(x) = p^{(0)}(x) + \hat{\lambda} p^{(1)}(x), \\ p_{\hat{\lambda}}(a) = 0, \\ \int_a^b p_{\hat{\lambda}} dx = 1. \end{cases} \quad (5.13)$$

From Eq. (5.13) it immediately follows that the conditions for $p^{(1)}$ are,

$$\begin{cases} p^{(1)}(a) = 0, \\ \int_a^b p^{(1)} dx = 0. \end{cases} \quad (5.14)$$

We can interpret Eq. (5.14) as requiring that the perturbed function $p_{\hat{\lambda}}$ maintain the absorbing boundary condition on the left and that no ‘mass’ is gained or lost. $p^{(1)}$ is defined implicitly by the eigenvalue problem for eigenvalue $\hat{\lambda}$ and eigenfunction $p_{\hat{\lambda}}$ about $\hat{\lambda} = 0$. Informally, we can think of the first-order correction $p^{(1)}$ as the ‘direction’ the solution of the eigenvalue problem is pushed as $\hat{\lambda}$ is perturbed. Given that $p_{\hat{\lambda}}$ is pinned at zero on the left boundary, the setting is reminiscent of a shooting problem, where we would be analysing how changing $\hat{\lambda}$ affects the path of our solution starting at $p_{\hat{\lambda}}(a) = 0$.

We now wish to solve for $p^{(1)}$ through its implicit eigenvalue problem defined as such,

$$\mathcal{L}_F p_{\hat{\lambda}} = \hat{\lambda} p_{\hat{\lambda}}.$$

By the definition of $p_{\hat{\lambda}}$ we have,

$$\mathcal{L}_F p^{(0)} + \hat{\lambda} \mathcal{L}_F p^{(1)} = \hat{\lambda} \left(p^{(0)} + \hat{\lambda} p^{(1)} \right).$$

By definition of $p^{(0)}$ and the fact that we are perturbing about $\hat{\lambda} = 0$, we arrive at,

$$\mathcal{L}_F p^{(1)} = p^{(0)}$$

That is to say, we wish to solve for $p^{(1)}$ in the following equation,

$$D \frac{\partial^2 p^{(1)}}{\partial x^2} + \frac{\partial}{\partial x} [U_x(x) p^{(1)}] = p^{(0)}.$$

Let $P^{(0)}(x) := \int_a^x p^{(0)}(s) ds$. Applying the integrating factor $e^{U(x)/D}$, integrating twice and applying the boundary and normalisation conditions in Eq. (5.14) gives the following solution in terms of integration constant c_1 ,

$$p^{(1)}(x) = \frac{1}{D} \int_a^x \exp \left(\frac{U(s) - U(x)}{D} \right) (P^{(0)}(s) + c_1) ds,$$

with constant c_1 given by,

$$c_1 = \frac{- \int_a^b \int_a^{s_2} \exp \left(\frac{U(s_1) - U(s_2)}{D} \right) P^{(0)}(s_1) ds_1 ds_2}{\int_a^b \int_a^{s_2} \exp \left(\frac{U(s_1) - U(s_2)}{D} \right) ds_1 ds_2}.$$

We now have expressions for $p^{(0)}$ and $p^{(1)}$, giving a complete picture of Eq. (5.13) with the free perturbation parameter $\hat{\lambda}$. The value of $\hat{\lambda}$ is determined by our choice of boundary condition at $x = b$, which until this point we have left undefined. We now explore values of $\hat{\lambda}$ which satisfy an absorbing or a homogeneous Neumann boundary condition at $x = b$.

Correction to absorbing boundary condition

Let us introduce an absorbing boundary condition $p_{\hat{\lambda}}(b) = 0$ to the perturbation approximation. Solving for the eigenvalue perturbation parameter $\hat{\lambda}$ in Eq. (5.13) gives

$$\hat{\lambda} = - \frac{p^{(0)}(b)}{p^{(1)}(b)}. \quad (5.15)$$

The corresponding perturbed solution $\hat{\psi}$ is given by

$$\hat{\psi}(x) = p^{(0)}(x) - \frac{p^{(0)}(b)}{p^{(1)}(b)} p^{(1)}(x). \quad (5.16)$$

Therefore, at $\hat{\lambda} = -\frac{p^{(0)}(b)}{p^{(1)}(b)}$ we have the approximated solution $\hat{\psi}$ of the eigenvalue problem $\mathcal{L}_F \hat{\psi} = \hat{\lambda} \hat{\psi}$, satisfying two boundary conditions and normalisation conditions. See Fig. 5.2 for an example of such an approximation being applied. That is, we have approximated solutions Eqs. (5.15) and (5.16) to the leading eigenvalue and eigenfunction of,

$$\begin{cases} \mathcal{L}_F \psi = \lambda \psi, & x \in [a, b], \\ \psi(a) = 0, \\ \psi(b) = 0, \\ \int_a^b \psi dx = 1. \end{cases} \quad (5.17)$$

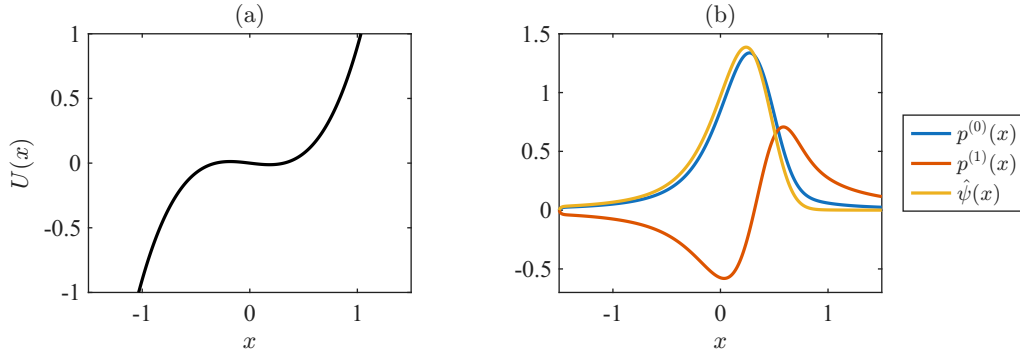


Figure 5.2: (a) An asymmetric potential function and (b) the corresponding BCPM approximation.

Correction to homogeneous Neumann boundary condition

If we wish to perturb our solution such that we have a homogeneous Neumann boundary condition at $x = b$ then a simple alteration is required from the method above. We wish to introduce the boundary condition $\frac{dp_{\hat{\lambda}}}{dx} = 0$ at $x = b$. Solving Eq. (5.13) gives,

$$\hat{\lambda} = -\frac{p_x^{(0)}(b)}{p_x^{(1)}(b)}, \quad (5.18)$$

with the corresponding perturbed solution $\hat{\psi}$ given by,

$$\hat{\psi}(x) = p^{(0)}(x) - \frac{p_x^{(0)}(b)}{p_x^{(1)}(b)} p^{(1)}(x), \quad (5.19)$$

where $p_x^{(0)} = \frac{dp^{(0)}}{dx}$ and $p_x^{(1)} = \frac{dp^{(1)}}{dx}$. However, as of yet, we have not computed the derivatives of $p^{(0)}$ or $p^{(1)}$. We can do so, giving the following,

$$\frac{dp^{(0)}}{dx} = c_0 - \frac{U_x(x)p^{(0)}(x)}{D},$$

and

$$\frac{dp^{(1)}}{dx} = \frac{c_1 + P^{(0)}(x) - U_x(x)p^{(1)}(x)}{D},$$

where c_0 , c_1 and $P^{(0)}(x)$ are defined as previously mentioned. Therefore we have an approximated solution Eqs. (5.18) and (5.19) to the leading eigenvalue and eigenfunction of,

$$\begin{cases} \mathcal{L}_F \psi = \lambda \psi, & x \in [a, b], \\ \psi(a) = 0, \\ \left. \frac{d\psi}{dx} \right|_{x=b} = 0, \\ \int_a^b \psi dx = 1. \end{cases} \quad (5.20)$$

Applying Neumann boundary condition to even anti-confining potential function

If we restrict the potential function $U(x)$ to even functions then one can gain additional approximations. Consider the FPO $\mathcal{L}_F$ with the the even anti-confining potential function $U(x)$ (that is $U(x) \rightarrow -\infty$ as $|x| \rightarrow \infty$) restricted to the domain $x \in [-\delta, \delta]$ with two absorbing boundary conditions. Assuming that U has a single, deep basin of attraction, by the even property of U , the stable equilibrium is $x^* = 0$. The standard application the BCPM, as presented above, fails in the case that $U(x) \rightarrow -\infty$ as $|x| \rightarrow \infty$ as we require an initially undefined boundary condition from which probability flux can enter the system at one boundary and exit at the other in a steady state manner. Therefore an alternative method is required.

As $U(x)$ is even, it follows that $\mathcal{L}_F$ commutes with the reflection operator $Rf(x) := f(-x)$. Therefore, if $\mathcal{L}_F$ has the eigenvalue λ_n with the corresponding eigenfunction $\psi_n(x)$, then $\psi_n(-x)$ must also be an eigenfunction of $\mathcal{L}_F$ (also corresponding to λ_n). Eigenfunctions for a given eigenvalue (in regular Sturm-Liouville problems) are unique up to a scalar multiple. Therefore $\psi_n(x) = c_n \psi_n(-x)$ for some scalar c_n . This can only hold for $c_n = \pm 1$, making

$\psi_n(x)$ either even or odd. Therefore we conclude that if $U(x)$ is an even function, then every eigenfunction of $\mathcal{L}_F$ is either even or odd.

By the Sturm oscillation theorem, ψ_n has n zeros in the open interval $(-\delta, \delta)$. An even function must have an even number of zeros, and an odd function must have an odd number of zeros. Therefore, if n is even, then ψ_n is even, and if n is odd, then ψ_n is odd. Therefore we can state the following,

$$\psi_{2k}(x) = \psi_{2k}(-x), \quad (5.21)$$

and

$$\psi_{2k+1}(x) = -\psi_{2k+1}(-x), \quad (5.22)$$

for $k = 0, 1, 2, \dots$ where $U(x)$ is an even potential function. At the origin, by the properties of even and odd functions, the following is also true,

$$\left. \frac{d\psi_{2k}}{dx} \right|_{x=0} = 0, \quad (5.23)$$

and

$$\psi_{2k+1}(0) = 0, \quad (5.24)$$

for $k = 0, 1, 2, \dots$

For some even potential function $U(x)$, the leading eigenfunction ψ_0 of $\mathcal{L}_F$ is therefore even with the condition $\left. \frac{d\psi_0}{dx} \right|_{x=0} = 0$. That is, we wish to solve the eigenvalue problem given by,

$$\begin{cases} \mathcal{L}_F \psi = \lambda \psi, & x \in [-\delta, 0], \\ \psi(-\delta) = 0, \\ \left. \frac{d\psi}{dx} \right|_{x=0} = 0, \\ \int_{-\delta}^0 \psi dx = 1. \end{cases} \quad (5.25)$$

These are exactly the conditions presented in Eq. (5.20) for the BCPM with a correction to the homogeneous Neumann boundary condition. We can therefore apply Eqs. (5.18) and (5.19) to the half domain of $[-\delta, 0]$ to find the approximated leading non-zero eigenvalue $\hat{\lambda}$ and eigenfunction $\hat{\psi}$. The half domain approximated eigenfunction given by Eq. (5.19) can naturally be extended to the full domain by the evenness property of ψ_0 . The eigenvalues on the half and full domain are equal. See Fig. 5.3 for the application of such an implementation of the BCPM.

Applying Neumann boundary condition to even confining potential functions

Consider the same FPO $\mathcal{L}_F$ with the the even anti-confining potential function $U(x)$ on the domain $x \in [-\delta, \delta]$ with two absorbing boundary conditions, as discussed in the section immediately prior. We established the approximations $\hat{\lambda}$ and $\hat{\psi}$ given by Eqs. (5.18) and (5.19) can

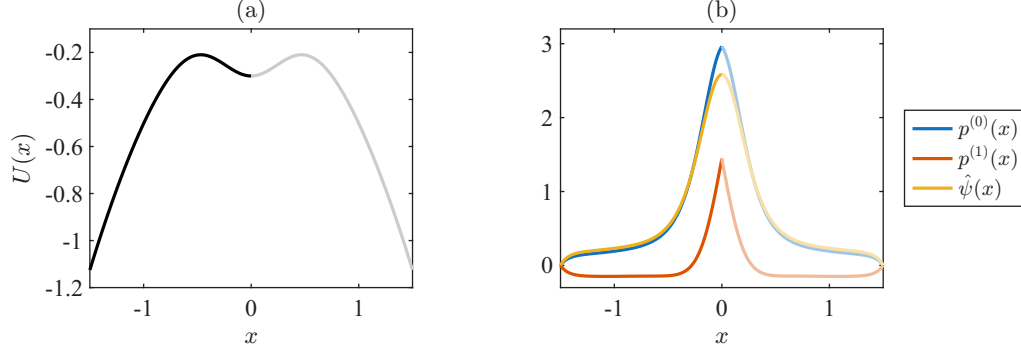


Figure 5.3: (a) An even anti-confining potential function, with the relevant half domain shaded darker and (b) the corresponding BCPM approximation applied to the half domain (shaded darker).

be applied. To make it clear that these approximations correspond to the leading eigenvalue and eigenfunction of $\mathcal{L}_F$, let $\hat{\lambda}_0 = \hat{\lambda}$ and $\hat{\psi}_0 = \hat{\psi}$. Given our approximations of $\mathcal{L}_F$, let us now consider the FPO $\bar{\mathcal{L}}_F$ with the inverted even potential function $\bar{U}(x) = -U(x)$ (which is therefore an even confining potential function). By Eq. (3.24) the corresponding boundary conditions for the inverted operator are two reflecting boundary conditions (i.e. two zero-flux boundary conditions) at $x = -\delta$ and $x = \delta$. Let $\bar{\mathcal{L}}_F$ have eigenvalues $\{\bar{\lambda}_0, \bar{\lambda}_1, \dots\}$ and eigenfunctions $\{\bar{\psi}_0, \bar{\psi}_1, \dots\}$.

Note that as $\bar{\mathcal{L}}_F$ has a two reflecting boundary conditions, the leading eigenvalue is $\bar{\lambda}_0 = 0$, the zero-eigenvalue. Therefore the leading non-zero eigenvalue is $\bar{\lambda}_1$. As stated in Eq. (3.27), we can relate eigenvalues of inverted potential functions as such,

$$\bar{\lambda}_1 = \lambda_0. \quad (5.26)$$

Therefore, the leading non-zero eigenvalue $\bar{\lambda}_1$ is approximated using the BCPM on the half domain $[-\delta, 0]$ with potential function $U(x)$, i.e. approximated by $\hat{\lambda}_0$. The leading non-zero eigenfunction $\bar{\psi}_1$ of $\bar{\mathcal{L}}_F$ can also be approximated from the transformation of $\hat{\psi}_0$, as presented in Eq. (3.27). However it is possible to compute an approximation of $\bar{\psi}_1$ directly within the BCPM framework, without the mentioned transformation, we shown this approach now.

As mentioned by Eq. (5.22), $\bar{\psi}_1$ is an odd function. By Eq. (5.24) we know $\bar{\psi}_1(0) = 0$. Let us once again consider the half domain, $x \in [0, \delta]$. Note that we use the positive half domain in this case where as the previous example used the negative half domain. The reason for the choice is simply to allow Eq. (5.13) to be applied directly in its present form as it choose the left boundary condition to satisfy $\bar{\psi}_1(0) = 0$ (a x to $-x$ variable change would allow the negative

half domain to be used if desired). Therefore we wish to approximate the following system,

$$\left\{ \begin{array}{l} \bar{\mathcal{L}}_F \bar{\psi} = \bar{\lambda} \bar{\psi}, \quad x \in [0, \delta], \\ \bar{\psi}(0) = 0, \\ J(\delta) = 0, \\ \int_0^\delta \bar{\psi} dx = 1, \end{array} \right. \quad (5.27)$$

where $J = -D \frac{d\bar{\psi}}{dx} - \frac{dU}{dx} \bar{\psi}$ is the probability flux associated with $\bar{\psi}$. We may begin our approximation in similar way to previous examples, with Eq. (5.13) giving,

$$\bar{p}_{\hat{\lambda}}(x) = \bar{p}^{(0)}(x) + \hat{\lambda} \bar{p}^{(1)}(x),$$

which satisfies the left absorbing boundary condition and the behaviour at the right boundary condition is dependent on $\hat{\lambda}$. In previous cases we were able to cleanly solve for the value of the eigenvalue perturbation parameter $\hat{\lambda}$ by enforcing a second boundary condition. Here, our second boundary condition is not so clean (a flux condition, rather than a homogeneous Dirichlet or Neumann condition), causing complexities of our expressions to grow quickly. However, by Eq. (5.26) we already know an approximation of λ from the anti-confining potential function case, therefore,

$$\hat{\lambda}_1 = \hat{\lambda}_0 = -\frac{p_x^{(0)}(0)}{p_x^{(1)}(0)},$$

where $p_x^{(0)}$ and $p_x^{(1)}(0)$ are the derivatives of correction functions from the even anti-confining potential function case (on the half domain). Therefore our approximation of leading non-zero eigenfunction $\hat{\psi}_1(x)$ of $\bar{\mathcal{L}}_F$ is given by,

$$\hat{\psi}_1(x) = \bar{p}_{\hat{\lambda}}(x) = \bar{p}^{(0)}(x) - \frac{p_x^{(0)}(0)}{p_x^{(1)}(0)} \bar{p}^{(1)}(x).$$

Once again, the eigenfunction approximation may be extended to the full domain by the oddness property of ψ_1 . See Fig. 5.4 for the application of such an implementation of the BCPM.

5.3 Rayleigh–Schrödinger perturbation method

We now present the *Rayleigh-Schrödinger perturbation method* (RSPM), a method widely used in quantum mechanics to approximate the eigenvalues and eigenfunctions of a differential operator given that a ‘similar’ self-adjoint operator has a known set of eigenvalues and eigenfunctions. Typically, the RSPM is applied to the Schrödinger operator. General notes for the

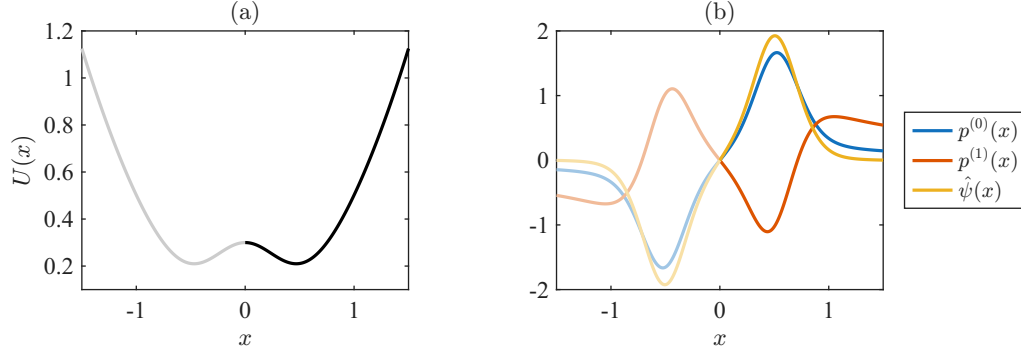


Figure 5.4: (a) A bistable even confining potential function, with the relevant half domain shaded darker and (b) the corresponding BCPM approximation applied to the half domain (shaded darker).

RSPM are adapted from Bender and Orszag [3] and then applied to the FPO. We also adapt the recursive formulation of the RSPM to allow for a more computationally efficient implementation to be applied in practice.

Let us decomposed the FPO $\mathcal{L}_F$ into a self-adjoint operator $\mathcal{L}_0$ with a know set of eigenvalues and eigenfunctions and another operator $\mathcal{L}_1$,

$$\mathcal{L}_F = \mathcal{L}_0 + \mathcal{L}_1. \quad (5.28)$$

Let $\mathcal{L}_0$ be self-adjoint with respect to the inner-product $\langle \cdot, \cdot \rangle$. Let us write the decomposition of $\mathcal{L}_F$ presented in Eq. (5.28) as a perturbation about $\mathcal{L}_0$ with perturbation parameter η as such,

$$\mathcal{L}_\eta = \mathcal{L}_0 + \eta \mathcal{L}_1. \quad (5.29)$$

We write the eigenvalues and eigenfunctions of $\mathcal{L}_\eta$ as $\{\lambda_0(\eta), \lambda_1(\eta), \dots\}$ and $\{\psi_0(x, \eta), \psi_1(x, \eta), \dots\}$, respectively. The eigenvalues and eigenfunctions of $\mathcal{L}_0$ are $\{\lambda_0^{(0)}, \lambda_1^{(0)}, \dots\}$ and $\{\psi_0^{(0)}(x), \psi_1^{(0)}(x), \dots\}$, respectively. We assume that the eigenfunctions $\{\psi_0^{(0)}(x), \psi_1^{(0)}(x), \dots\}$ form an orthonormal basis of our given Hilbert space. Each $\lambda_n(\eta)$ and $\psi_n(x, \eta)$ can be written as a perturbation as such,

$$\lambda_n(\eta) = \lambda_n^{(0)} + \eta \lambda_n^{(1)} + \eta^2 \lambda_n^{(2)} + \dots, \quad (5.30)$$

$$\psi_n(\eta, x) = \psi_n^{(0)}(x) + \eta \psi_n^{(1)}(x) + \eta^2 \psi_n^{(2)}(x) + \dots \quad (5.31)$$

We call $\lambda_n^{(m)}$ and $\psi_n^{(m)}(x)$ the m th order correction to the n th eigenvalue and eigenfunction, respectively. We wish to solve for correction terms in Eqs. (5.30) and (5.31) such that $\lambda_n(\eta)$ and $\psi_n(x, \eta)$ satisfy the eigenvalue equation of $\mathcal{L}_\eta$,

$$\mathcal{L}_\eta \psi_n(x, \eta) = \lambda_n(\eta) \psi_n(x, \eta).$$

Substituting Eqs. (5.29) to (5.31),

$$(\mathcal{L}_0 + \eta \mathcal{L}_1)[\psi_n^{(0)} + \eta \psi_n^{(1)} + \eta^2 \psi_n^{(2)} + \dots] = (\lambda_n^{(0)} + \eta \lambda_n^{(1)} + \eta^2 \lambda_n^{(2)} + \dots)(\psi_n^{(0)} + \eta \psi_n^{(1)} + \eta^2 \psi_n^{(2)} + \dots).$$

Expanding and grouping by orders of η ,

$$\eta^0 : \quad \mathcal{L}_0 \psi_n^{(0)} = \lambda_n^{(0)} \psi_n^{(0)}, \quad (5.32)$$

$$\eta^1 : \quad (\mathcal{L}_0 - \lambda_n^{(0)}) \psi_n^{(1)} = -\mathcal{L}_1 \psi_n^{(0)} + \lambda_n^{(1)} \psi_n^{(0)}, \quad (5.33)$$

$$\eta^2 : \quad (\mathcal{L}_0 - \lambda_n^{(0)}) \psi_n^{(2)} = -\mathcal{L}_1 \psi_n^{(1)} + \lambda_n^{(1)} \psi_n^{(1)} + \lambda_n^{(2)} \psi_n^{(0)}, \quad (5.34)$$

$$\vdots$$

$$\eta^m : \quad (\mathcal{L}_0 - \lambda_n^{(0)}) \psi_n^{(m)} = -\mathcal{L}_1 \psi_n^{(m-1)} + \sum_{i=1}^m \lambda_n^{(i)} \psi_n^{(m-i)}. \quad (5.35)$$

By assumption, we take Eq. (5.32) to be true with known solutions. Higher orders of η are used to solve for correction terms. We can express the m th order corrections $\psi_n^{(m)}$ using the orthonormal basis $\{\psi_0^{(0)}, \psi_1^{(0)}, \psi_2^{(0)}, \dots\}$ with coefficients $\{a_{n,0}^{(m)}, a_{n,1}^{(m)}, a_{n,2}^{(m)}, \dots\}$ as such,

$$\psi_n^{(m)} = \sum_i a_{n,i}^{(m)} \psi_i^{(0)} \quad (5.36)$$

This expression of higher order corrections will be used extensively throughout the following derivations. Solutions for the first-order and second-order corrections found in Eqs. (5.33) and (5.34) will now be presented, followed by a general solution for the m th order corrections.

First order corrections

Substituting Eq. (5.36) into Eq. (5.33) and utilising linearity and Eq. (5.32),

$$\sum_i a_{n,i}^{(1)} (\lambda_i^{(0)} - \lambda_n^{(0)}) \psi_i^{(0)} = -\mathcal{L}_1 \psi_n^{(0)} + \lambda_n^{(1)} \psi_n^{(0)}.$$

Applying $\langle \psi_j^{(0)}, \cdot \rangle$,

$$\sum_i a_{n,i}^{(1)} (\lambda_i^{(0)} - \lambda_n^{(0)}) \langle \psi_j^{(0)}, \psi_i^{(0)} \rangle = -\langle \psi_j^{(0)}, \mathcal{L}_1 \psi_n^{(0)} \rangle + \lambda_n^{(1)} \langle \psi_j^{(0)}, \psi_n^{(0)} \rangle. \quad (5.37)$$

Case $j = n$. As $\psi_n^{(0)}$ and $\psi_i^{(0)}$ are orthonormal for $n \neq i$ we can drop all terms in the summation on the LHS of Eq. (5.37) except for $n = i$. However, when $n = i$ the term in the summation is equal to zero due to $\lambda_i^{(0)} - \lambda_n^{(0)}$. Thus in the case $j = n$, Eq. (5.37) yields the first order eigenvalue correction,

$$\lambda_n^{(1)} = \langle \psi_n^{(0)}, \mathcal{L}_1 \psi_n^{(0)} \rangle \quad (5.38)$$

Case $j \neq n$. Once again, by orthogonality, several terms drop out of Eq. (5.37). The only term left in the summation on the LHS of Eq. (5.37) is from $i = j$, leaving $(\lambda_j^{(0)} - \lambda_n^{(0)})a_{n,j}^{(1)}$. Rearranging gives,

$$a_{n,j}^{(1)} = \frac{-\langle \psi_j^{(0)}, \mathcal{L}_1 \psi_n^{(0)} \rangle}{\lambda_j^{(0)} - \lambda_n^{(0)}}, \quad j \neq n.$$

Note that $a_{n,n}^{(m)}$ is equal to zero as for all n and m (a process called the *intermediate normalization* in some literature). We can think of this as perturbing away from $\psi_n^{(0)}$ for all n , hence there is no contribution from $\psi_n^{(0)}$ when expressing each correction term (hence $a_{n,n}^{(m)} = 0$). Substituting into Eq. (5.36) yields the first order eigenfunction correction,

$$\psi_n^{(1)} = \sum_{i \neq n} \frac{-\langle \psi_i^{(0)}, \mathcal{L}_1 \psi_n^{(0)} \rangle}{\lambda_i^{(0)} - \lambda_n^{(0)}} \psi_i^{(0)}. \quad (5.39)$$

Second order corrections

Let us apply a similar treatment to Eq. (5.34) to yield the second order corrections. Substituting Eq. (5.36) into Eq. (5.34),

$$\sum_i a_{n,i}^{(2)} (\lambda_i^{(0)} - \lambda_n^{(0)}) \psi_i^{(0)} = -\mathcal{L}_1 \psi_n^{(1)} + \lambda_n^{(1)} \psi_n^{(1)} + \lambda_n^{(2)} \psi_n^{(0)}.$$

Applying $\langle \psi_j^{(0)}, \cdot \rangle$,

$$\sum_i a_{n,i}^{(2)} (\lambda_i^{(0)} - \lambda_n^{(0)}) \langle \psi_j^{(0)}, \psi_i^{(0)} \rangle = -\langle \psi_j^{(0)}, \mathcal{L}_1 \psi_n^{(1)} \rangle + \lambda_n^{(1)} \langle \psi_j^{(0)}, \psi_n^{(1)} \rangle + \lambda_n^{(2)} \langle \psi_j^{(0)}, \psi_n^{(0)} \rangle.$$

Case $j = n$. Applying the same orthogonality practices as the first order correction as well as utilising the orthogonality of $\psi_n^{(0)}$ and $\psi_n^{(m)}$ for all $m \geq 1$.

$$\lambda_n^{(2)} = \langle \psi_n^{(0)}, \mathcal{L}_1 \psi_n^{(1)} \rangle.$$

Case $j \neq n$. Following same logic as the first order correction,

$$a_{n,j}^{(2)} = \frac{-\langle \psi_j^{(0)}, \mathcal{L}_1 \psi_n^{(1)} \rangle + \lambda_n^{(1)} \langle \psi_j^{(0)}, \psi_n^{(1)} \rangle}{\lambda_j^{(0)} - \lambda_n^{(0)}}, \quad j \neq n.$$

Substituting into Eq. (5.36),

$$\psi_n^{(2)} = \sum_{i \neq n} \frac{-\langle \psi_i^{(0)}, \mathcal{L}_1 \psi_n^{(1)} \rangle + \lambda_n^{(1)} \langle \psi_i^{(0)}, \psi_n^{(1)} \rangle}{\lambda_i^{(0)} - \lambda_n^{(0)}} \psi_i^{(0)}. \quad (5.40)$$

General higher order corrections

Let us work more generally and find expressions for the m th order corrections $\lambda_n^{(m)}$ and $\psi_n^{(m)}$ found in Eq. (5.35). The logic is almost identical to that used to find the first and second order corrections. We begin by substituting Eq. (5.36) into Eq. (5.35) and applying linearity,

$$\sum_i a_{n,i}^{(m)} (\lambda_i^{(0)} - \lambda_n^{(0)}) \psi_i^{(0)} = -\mathcal{L}_1 \psi_n^{(m-1)} + \sum_{i=1}^m \lambda_n^{(i)} \psi_n^{(m-i)}.$$

Applying $\langle \psi_j^{(0)}, \cdot \rangle$,

$$\sum_i a_{n,i}^{(m)} (\lambda_i^{(0)} - \lambda_n^{(0)}) \langle \psi_j^{(0)}, \psi_i^{(0)} \rangle = -\langle \psi_j^{(0)}, \mathcal{L}_1 \psi_n^{(m-1)} \rangle + \sum_{i=1}^m \lambda_n^{(i)} \langle \psi_j^{(0)}, \psi_n^{(m-i)} \rangle.$$

Case $j = n$. Apply normal orthogonality tricks on the LHS and apply the orthogonality of $\psi_n^{(0)}$ and $\psi_n^{(m)}$ for all $m \geq 1$ on the RHS, reducing the summation to simply $\lambda_n^{(m)}$.

$$\lambda_n^{(m)} = \langle \psi_n^{(0)}, \mathcal{L}_1 \psi_n^{(m-1)} \rangle.$$

This is a clean, general expression for the m th order eigenvalue corrections. However, when actually computing this inner-product it may be more useful to write in terms of the known basis $\{\psi_0^{(0)}, \psi_1^{(0)}, \dots\}$ by substituting Eq. (5.36) and utilising linearity as such,

$$\lambda_n^{(m)} = \sum_i a_{n,i}^{(m-1)} \langle \psi_n^{(0)}, \mathcal{L}_1 \psi_i^{(0)} \rangle.$$

Case $j \neq n$. Once again, orthogonality leaves $a_{n,j}^{(m)} (\lambda_j^{(0)} - \lambda_n^{(0)})$ on the LHS. The summation on the RHS remains unchanged apart from the term corresponding to $i = m$ which is removed by orthogonality, causing the summation to terminate at $i = m - 1$ (rather than $i = m$ as before).

$$a_{n,j}^{(m)} = \frac{-\langle \psi_j^{(0)}, \mathcal{L}_1 \psi_n^{(m-1)} \rangle + \sum_{i=1}^{m-1} \lambda_n^{(i)} \langle \psi_j^{(0)}, \psi_n^{(m-i)} \rangle}{\lambda_j^{(0)} - \lambda_n^{(0)}}. \quad (5.41)$$

Once again, we could simply substitute Eq. (5.41) into Eq. (5.36) and we would have an acceptable expression for $\psi_n^{(m)}$ however we wish to express the corrections in terms of the known basis $\{\psi_0^{(0)}, \psi_1^{(0)}, \dots\}$. Let us consider each term of Eq. (5.41) written in terms of $\{\psi_0^{(0)}, \psi_1^{(0)}, \dots\}$ and simplify. Firstly,

$$\begin{aligned} \langle \psi_j^{(0)}, \mathcal{L}_1 \psi_n^{(m-1)} \rangle &= \langle \psi_j^{(0)}, \mathcal{L}_1 \left[\sum_i a_{n,i}^{(m-1)} \psi_i^{(0)} \right] \rangle, \\ &= \sum_i a_{n,i}^{(m-1)} \langle \psi_j^{(0)}, \mathcal{L}_1 \psi_i^{(0)} \rangle. \end{aligned} \quad (5.42)$$

Next,

$$\begin{aligned}
 \sum_{i=1}^{m-1} \lambda_n^{(i)} \langle \psi_j^{(0)}, \psi_n^{(m-i)} \rangle &= \sum_{i=1}^{m-1} \lambda_n^{(i)} \langle \psi_j^{(0)}, \sum_k a_{n,k}^{(m-i)} \psi_k^{(0)} \rangle, \\
 &= \sum_{i=1}^{m-1} \lambda_n^{(i)} \sum_k a_{n,k}^{(m-i)} \langle \psi_j^{(0)}, \psi_k^{(0)} \rangle, \\
 &= \sum_{i=1}^{m-1} a_{n,j}^{(m-i)} \lambda_n^{(i)}.
 \end{aligned} \tag{5.43}$$

Substituting Eqs. (5.42) and (5.43) into Eq. (5.41) gives $a_{n,j}^{(m)}$ in terms of the known basis $\{\psi_0^{(0)}, \psi_1^{(0)}, \dots\}$ as such,

$$a_{n,j}^{(m)} = \frac{-\sum_i a_{n,i}^{(m-1)} \langle \psi_j^{(0)}, \mathcal{L}_1 \psi_i^{(0)} \rangle + \sum_{i=1}^{m-1} a_{n,j}^{(m-i)} \lambda_n^{(i)}}{\lambda_j^{(0)} - \lambda_n^{(0)}}.$$

Therefore, any order of correction of the RSPM can be expressed in terms of the known basis $\{\psi_0^{(0)}, \psi_1^{(0)}, \dots\}$.

Chapter 6

Results

We now wish to apply and compare the perturbation methods discussed in previous chapters to concrete examples. We compare the BCPM with the RSPM in the case of a bistable confining potential function.

In Sec. 6.1 the experimental settings under which experiments are conducted are covered. Sec. 6.2 presents the results for approximations of the eigenvalues and eigenfunctions of the FPO. In Sec. 6.3 empirical results tipping rate results are compared with perturbation approximations. Finally, Sec. 6.4 presents results for time-dependent eigenmode approximations.

6.1 Experimental setting

Given the limitations of potential functions to which the RSPM and the BCPM can be applied, the results in the following section are produced for a setting in which both methods are applicable, allowing comparisons to be made. For the entirety of this chapter we study the FPO $\mathcal{L}_F$ with the even bistable potential function U with bifurcation parameter μ , and parameters α and β , where U is defined as such,

$$U(x, \mu(t)) = \frac{\alpha}{2}x^2 + \left(\frac{\alpha}{\beta} - \mu\right)e^{-\frac{\beta}{2}x^2}. \quad (6.1)$$

Eq. (6.1) is simply a parabolic potential function perturbed by a Gaussian-type bump. The bifurcation parameter $\mu(t)$ is defined to produce a bifurcation at $\mu^* = 0$ and $x^* = 0$. We observe the bifurcation by ramping $\mu(t)$ from a negative value towards 0. The parameter α determines the slope of the parabolic bowl of the potential function, and β determines the slope of the Gaussian bump centred at $x = 0$. Eq. (6.1) yields an even confining potential function with bistability and a natural decomposition into an operator with a known set of eigenvalues and eigenfunctions. This decomposition of U is given by,

$$U(x, \mu(t)) = U_0(x) + \eta U_1(x), \quad (6.2)$$

where,

$$U_0(x) = \frac{\alpha}{2}x^2, \quad (6.3)$$

$$U_1(x) = e^{-\frac{\beta}{2}x^2}, \quad (6.4)$$

and the perturbation parameter η maps to the bifurcation parameter μ as such,

$$\eta = \frac{\alpha}{\beta} - \mu.$$

As described in Sec. 5.1, U_0 is the parabolic potential and offers a known set of eigenvalues and eigenfunctions, a requirement for the RSPM to be applied. In addition, U_1 is ‘relatively small’ compared to U_0 in the sense that U_1 decays away very quickly, producing a relatively localised perturbation. The resulting decomposition of $\mathcal{L}_F$ is given by,

$$\mathcal{L}_F = \mathcal{L}_0 + \eta\mathcal{L}_1, \quad (6.5)$$

where

$$\mathcal{L}_0 p = D \frac{\partial^2 p}{\partial x^2} + \frac{\partial}{\partial x} \left[\frac{\partial U_0}{\partial x} p \right],$$

and

$$\mathcal{L}_1 p = \frac{\partial}{\partial x} \left[\frac{\partial U_1}{\partial x} p \right].$$

For the remainder of this chapter, we choose $\alpha = 1$ and $\beta = 10$ as a case study. The choices of α and β have no significant qualitative effect on the results. Given this selection of parameters, the bifurcation of U in terms of η occurs at $\eta^* = \frac{\alpha}{\beta} - \mu^* = 0.1$. Visualisations of U_0 , U_1 and U (for varied η) are shown in Fig. 6.1.

Three distinct experiments are carried out in this chapter. For all three experiments, we range the diffusion constant D from 10^{-2} to 10^1 and range the bifurcation parameter μ from -1 to 0 (the corresponding range of the perturbation parameter η is 0.1 to 1.1). These parameter ranges are sufficiently large to allow for diverse cases to be studied. Diffusion constants lower than 10^{-2} begin to yield ‘quasi-deterministic’ results, with tipping occurring at very low rates prior to the bifurcation. In Secs. 6.2 and 6.3 each combination of D and μ is tested in isolation. That is to say, there is no ramping of $\mu(t)$ over time and we simply study $\mathcal{L}_F$ for given values of μ and D . Sec. 6.2 presents the results for eigenvalue approximations of the RSPM and BCPM. Sec. 6.3 assesses the ability of each method to approximate the empirical tipping rate between potential wells for given values of μ and D by using the leading non-zero eigenvalue. In contrast, Sec. 6.4 explores temporal eigenmode approximations of solutions of the FPE. During Sec. 6.4 the bifurcation parameter $\mu(t)$ is varied at a different of rates ε , with $\mu(t)$ given by,

$$\mu(t) = -1 + \varepsilon t$$

for $t \in [0, 1/\varepsilon]$. That is, we ramp $\mu(t)$ from -1 to 0 in a linear manner. We evaluate ε values from 10^{-4} to 10^0 .

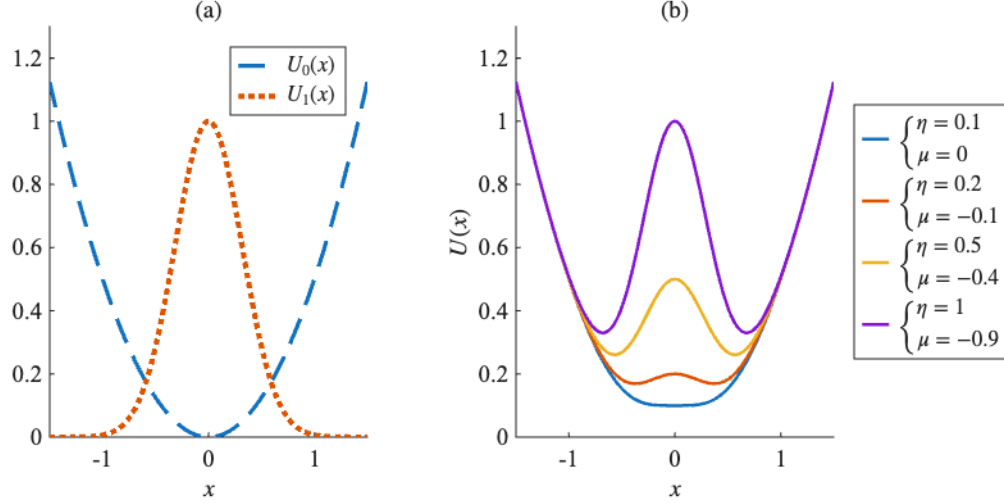


Figure 6.1: (a) Potential function U_0 given by Eq. (6.3) with $\alpha = 1$ and potential function U_1 given by Eq. (6.4) with $\beta = 10$. (b) Potential function U given by Eq. (6.2) with varied perturbation parameter η and the equivalent bifurcation parameter μ .

Numerical validation

To validate the eigenvalue, eigenfunction and time-dependent eigenmode approximations we employ numerical methods. In Secs. 6.2 to 6.4, all ‘errors’ are the evaluated by treating the following numerical methods as exactly correct results. All numerical methods are implemented using MATLAB [12].

In Sec. 6.2, perturbation approximations of eigenvalues and eigenfunctions of the FPO are validated using numerical results produced by the *Chebfun* package in MATLAB [10]. Chebfun uses piecewise polynomial interpolation to approximate continuous functions and differential operators to machine precision (approximately 15 digits of relative accuracy for functions and operators, yielding approximately 10 digits of relative accuracy for eigenvalue approximations). A restriction to a finite domain is required to apply Chebfun. This domain must include all dynamics of interest and be sufficiently large to allow for significant decaying of tails of eigenfunctions (see Eq. (4.4)). We do so by computing domain $[a, b]$ for which the tails of the steady state distribution have decayed to at least 10^{-12} , and then boundary conditions are applied as described in Sec. 3.1.

When numerically approximating SDE realisations in Sec. 6.3 we turn to the classical *Euler-Maruyama method* (see [14]). The method uses a time discretisation $\{t_0, t_1, \dots, t_N\}$, with approximated discrete values of $X(t)$ given as $\{\hat{X}(t_0), \hat{X}(t_1), \dots, \hat{X}(t_N)\}$ and an initial condition $\hat{X}(t_0) = x_0$. Let us assume a uniform time discretisation, giving a time step size of

$\Delta t = t_{n+1} - t_n$. The evolution of $\hat{X}$ is given by the iterative scheme,

$$\hat{X}(t_{n+1}) = f(\hat{X}(t_n), \mu(t_n))\Delta t \hat{X}(t_n) + \sigma\sqrt{\Delta t} Y_n,$$

where $\{Y_0, Y_1, \dots, Y_N\}$ are independent, identically distributed random variables from the standard normal distribution (i.e Gaussian distribution with mean 0 and variance 1).

In Sec. 6.4, we validate the approximated solutions to the FPE by applying the standard `pdepe` solver function, native to MATLAB [26].

6.2 Approximations of eigenvalues and eigenfunctions

Given the experimental setting described in Sec. 6.1, we now compare the ability of the RSPM and the BCPM to approximate the eigenvalues and eigenfunctions of $\mathcal{L}_F$ under varied parameters. The primary point of comparison will be the approximation of leading non-zero eigenvalue. The leading non-zero eigenvalue is of particular interest due to its use as an indicator of tipping rate, for example Eqs. (4.19) and (4.20). Furthermore, the BCPM is limited to approximations of the leading non-zero eigenvalue. As our experiments analyse a confining potential function described by Eq. (6.1), the first eigenvalue of $\mathcal{L}_F$ is zero, that is, $\lambda_0 = 0$. Therefore, the leading non-zero eigenvalue in this experiment is λ_1 .

Convergence of Rayleigh-Schrödinger perturbation method

We briefly cover the convergence of the RSPM. When applying the RSPM in practice, two truncations must be made, namely the highest order of correction m and the number of basis functions n . The effect of each of these parameters on errors of leading non-zero eigenvalue approximations are shown in Fig. 6.2.

As one might expect, increasing either n or m results in an error reduction in most cases. An exception is in the case of divergent series, in which increasing m may cause the error to increase. This case is exemplified by the plot for $\eta = 1$ in Fig. 6.2a, where errors diverge as m is increased. It seems that there is some radius of convergence in terms of η for which the RSPM converges. Furthermore, in Fig. 6.2a we can empirically see that values of η which appear to be converging, do so at different rates, with smaller η values showing faster convergence (in terms of m).

For the remainder of this chapter, we choose the number of basis functions $n = 10$ and the highest order of correction $m = 3$. The selection of a relatively low highest order of correction is partially motivated by the fact that the BCPM, with which we wish to compare the RSPM, is a first order method by design. In addition, we desire simplifications where possible and, as such, very high orders of correction should be avoided.

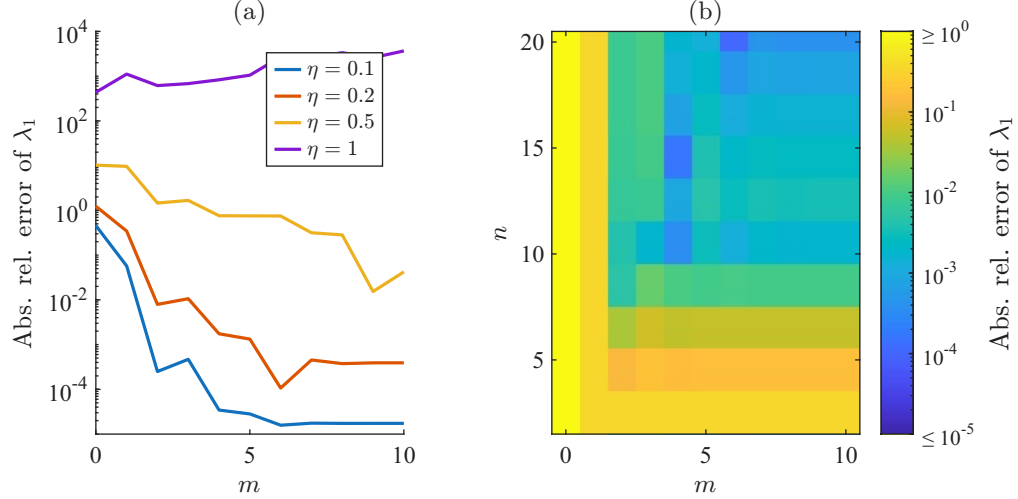


Figure 6.2: (a) Convergence of RSPM with varied order of correction m , varied perturbation parameter η , fixed number of basis functions $n = 20$ and fixed diffusion constant $D = 0.1$. (b) Convergence of RSPM with varied order of correction m , varied number of basis functions n , fixed perturbation parameter $\eta = 0.2$ and fixed diffusion constant $D = 0.1$. The line for $\eta = 0.2$ in (a) corresponds with the row at $n = 20$ in (b).

Comparison of methods

The RSPM and BCPM are applied to a range of parameters, for each of which an approximation of the eigenvalue(s) and eigenfunction(s) of $\mathcal{L}_F$ are made. Errors for leading non-zero eigenvalue approximations are presented in Fig. 6.3. A summary of the regions of the parameter space for which each approximation method offers best accuracy is presented in Fig. 6.4. In addition to the leading non-zero eigenvalue, the RSPM is also capable of approximating higher eigenvalues and eigenfunctions. The errors for the second, third, fourth and fifth eigenvalues are presented in Fig. 6.5.

In Figs. 6.3 to 6.5, no results are shown for very low diffusion constant D and very negative bifurcation parameter μ . In this region the leading non-zero eigenvalue is so small that $\lambda_1 \approx 0$ up to machine precision (limited to approximately ten decimal places with the solvers used in these experiments). Therefore our validation method of applying the Chebfun package is not available.

Immediately it is clear that the RSPM and BCPM each perform better in different parameter ranges. The BCPM thrives in low noise settings, far from the bifurcation point. This is exactly the portion of the parameter space in which the potential well is very deep relative to the noise level. This result is as expected as the BCPM perturbs away from a case in which the leading non-zero eigenvalue is assumed to be zero (which is almost true for deep potential

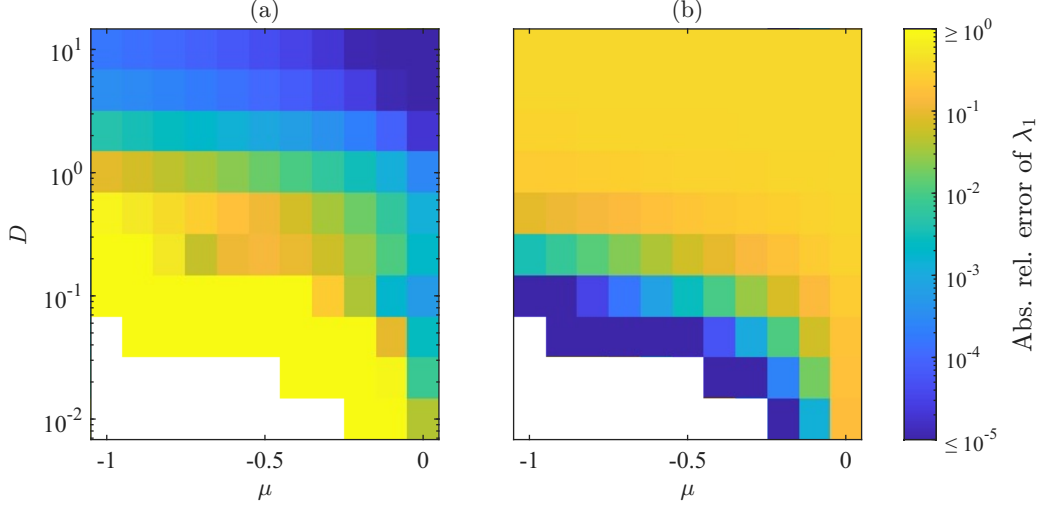


Figure 6.3: The absolute relative errors of leading non-zero eigenvalue approximations of (a) the RSPM with fixed number of basis functions $n = 10$ and fixed order of correction $m = 3$ and (b) the BCPM. Each point in the grid of errors is the error for a given diffusion constant D and bifurcation parameter μ .

wells relative to the noise level). Although the validation methods in this thesis did not have sufficient numerical precision to comfortably make claims about the accuracy of the BCPM for extremely deep potential wells relative to the noise, one might assume that the relative accuracy would only continue to improve as the noise continues to decrease.

In contrast the RSPM performs much better in either high noise settings, or μ values close to/at the bifurcation point. This can be explained by considering Eq. (6.5). As the RSPM is perturbing from $\mathcal{L}_0$ by $\eta\mathcal{L}_1$, we expect high accuracy when the ‘size’ of $\eta\mathcal{L}_1$ is small relative to the ‘size’ of $\mathcal{L}_0$. This is clearly the case when $|\eta|$ is close to zero. As $\eta = 0.1 - \mu$ in the experimental setting described in Sec. 6.1, $|\eta|$ is smallest for $\mu \approx 0.1$, with $|\eta|$ decreasing as μ approaches the bifurcation point $\mu^* = 0$ from the left (from the negatives). This explains the good accuracy of the RSPM for μ values close to/at the bifurcation point. In high noise settings we also see that $\eta\mathcal{L}_1$ is small relative to $\mathcal{L}_0$ because $\mathcal{L}_0$ has a noise/diffusion term $D\frac{\partial^2}{\partial x^2}$ (as well as a drift term), where as $\mathcal{L}_1$ has only a drift term. Therefore, when D is dominant, so is $\mathcal{L}_0$.

Although the RSPM has partially addressed the issue of the limited parameter space for which the BCPM is accurate, there is an intermediate regime which is not well approximated by either the RSPM or the BCPM, as shown in Fig. 6.4.

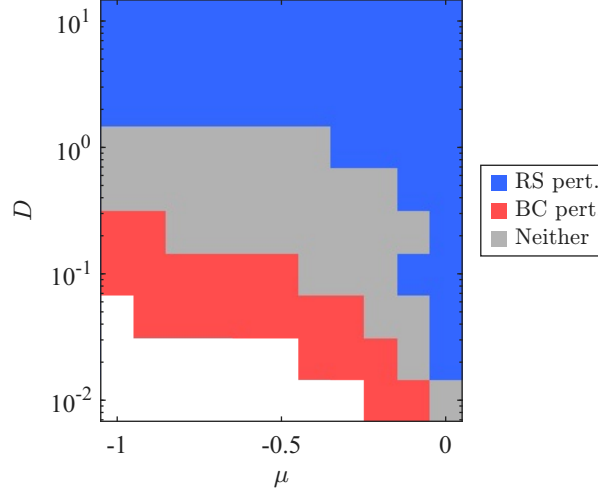


Figure 6.4: Regions of diffusion constant D and bifurcation parameter μ parameter space for which the RSPM or BCPM (or neither) yield absolute relative errors of leading non-zero eigenvalue less than 10^{-2} .

6.3 Empirical tipping rates

Once again working under the experimental setting presented in Sec. 6.1, for a given diffusion constant D and bifurcation parameter μ we simulate a realisation of the SDE Eq. (3.1) with initial condition $X(0) = x_R^*(0)$, where x_R^* is the right hand stable equilibrium. See Fig. 6.6 for an example of such a realisation.

Let us consider a realisation $X(t)$ of Eq. (3.1) for $t \in [0, T]$. Let $N(t)$ be the number of transitions from one stable equilibrium to another at time t for a given realisation of $X(t)$ (i.e. number of tipping events). Given that a realisation is allowed to run for a sufficiently long period of time (i.e. a sufficiently large T), the empirical estimated tipping rate $\hat{r}_{emp}$ can be computed as such,

$$\hat{r}_{emp} = \frac{N(T)}{T}.$$

Recall from Eq. (4.20) that the leading non-zero eigenvalue also approximates tipping rate. Let this eigenvalue estimate be given by,

$$\hat{r}_\lambda = \frac{-\lambda_1}{2}.$$

Let us find the sample size T for which the sample size is large such that the empirical tipping rate is an accurate estimate of the ‘true’ tipping rate r . We want the probability of the absolute relative error $|\Delta r|$ being within some tolerance ϵ with probability ρ . That is, we

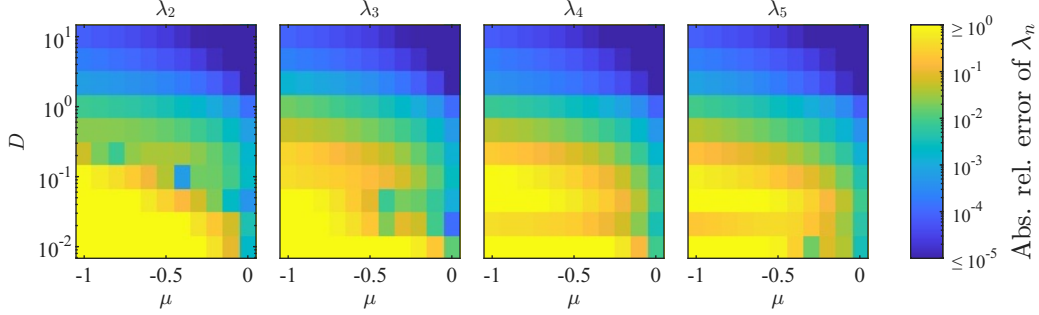


Figure 6.5: The absolute relative errors of higher non-zero eigenvalue approximations of the RSPM with fixed number of basis functions $n = 10$ and fixed order of correction $m = 3$.

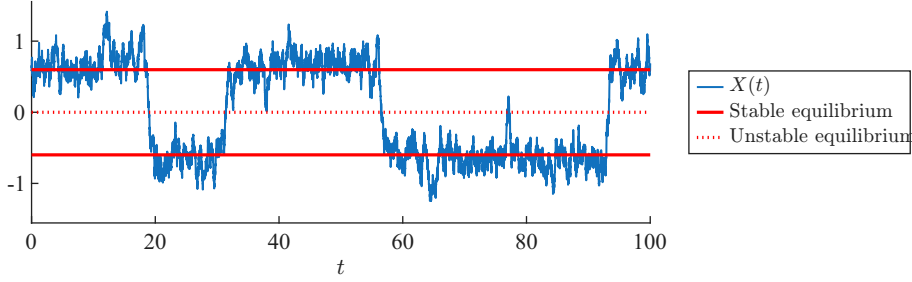


Figure 6.6: A realisation of the SDE Eq. (3.1) for fixed $D = 0.1$ and $\mu = -0.5$ in the bistable potential function given in Eq. (6.1).

desire,

$$P(|\Delta r| \leq \epsilon) \geq \rho,$$

where

$$\Delta r = \frac{\hat{r}_{emp} - r}{r}.$$

We must estimate the distribution of our empirical error Δr . Let us assume that $N(t)$ is distributed by Poisson point process with rate r , that is, we assume $N(t) \sim \text{Pois}(rt)$. Assuming a sufficient large T , by the central limit theorem we can approximately write,

$$\frac{\sqrt{T}}{\sqrt{r}}(\hat{r}_{emp} - r) \sim \mathcal{N}(0, 1),$$

where $\mathcal{N}$ is the normal distribution. Therefore, the relative error Δr of $\hat{r}_{emp}$ is given by

$$\Delta r = \frac{\hat{r}_{emp} - r}{r} \sim \mathcal{N}(0, 1/rT).$$

Applying the inverse normal distribution and solving for T yields,

$$T \geq \frac{z(\rho)^2}{\epsilon^2 r}, \quad (6.6)$$

where $z(x) = \Phi^{-1}((1+x)/2)$ is a transformation of inverse cumulative standard normal distribution Φ^{-1} . That is, a realisation of $X(t)$ must be simulated for a time duration of at least $\frac{z(\rho)^2}{\epsilon^2 r}$ to achieve an empirical tipping rate with a relative error less than or equal to ϵ with a probability of at least ρ . We choose, $\epsilon = 0.05$ and $\rho = 0.95$ for our experiments and determine the duration T of a realisation of $X(t)$ as such.

In Fig. 6.7 we present the relative error of approximating $\hat{r}_{emp}$ using $\hat{r}_\lambda$, with comparisons of the λ_1 values produced by the BCPM, the RSPM and the Chebfun package. We restrict this experiment to $T \leq 10^7$ for computational reasons. Thus, as seen in Fig. 6.7, if the T required to satisfy Eq. (6.6) exceeds 10^7 , we leave those parameters blank.

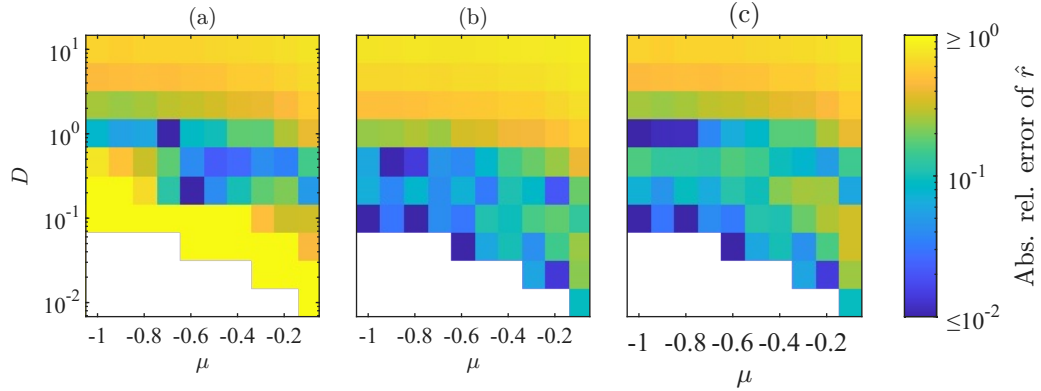


Figure 6.7: Absolute relative errors of leading non-eigenvalue λ_1 used as an estimation of empirical tipping rate $\hat{r}$ for a given diffusion constant D and bifurcation parameter μ . The value of λ_1 used is (a) the RSPM, (b) the BCPM, and (c) the Chebfun package.

The results presented in Fig. 6.7 have variation of errors, even between similar parameters, due to the empirical natural of the experiment. For low noise settings the variation is approximately in line with our tolerances. The use of the leading eigenvalue to estimate tipping rates assumes a potential function with two deep potential wells relative to the strength of noise. Therefore it is expected that we see high relative errors (even when using the Chebfun package to estimate λ_1) in high noise regions in Fig. 6.7. If we compare Fig. 6.7b and Fig. 6.7c, the region for which the BCPM well approximates the leading non-zero eigenvalue does not appear too much smaller than region for which the leading non-zero eigenvalue well approximates tipping rates.

6.4 Eigenmode approximations

We now present results for the application of eigenmode approximations to the FPE Eq. (3.2), with ramped, time-dependent bifurcation parameter $\mu(t)$. Once again, we are operating under the experimental setting described in Sec. 6.1. In all experiments, the initial condition of the PDF is given by $p(x, 0) = \delta(x - x_R^*(0))$ where x_R^* is the right hand stable equilibrium. That is to say, all experiments begin in the right hand basin of attraction. We then ramp the bifurcation parameter $\mu(t)$ at rate ε from -1 to 0 . As the bifurcation parameter is ramped, the basin of attraction shallows, increasing the rate at which transitions between basins may occur. We track the probability of being in the right hand basin with $p_R(t)$, given by,

$$p_R(t) = \int_0^\infty p(x, t) dx.$$

Initially $p_R(0) = 1$, due to our initial condition for $p(x, 0)$. The symmetry of our system gives $p_R = 0.5$ in the steady state distribution. Therefore we expect to see a transition from $p_R = 1$ to $p_R = 0.5$ over time. This can be seen in Fig. 6.8.

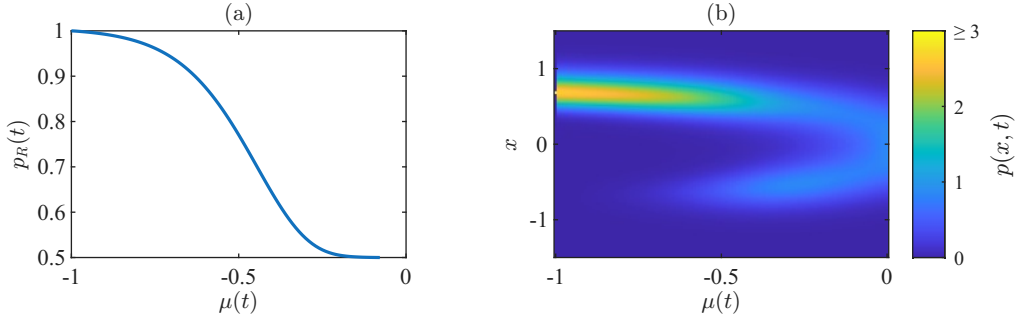


Figure 6.8: For a fixed diffusion constant $D = 10^{-1}$ and parameter rate of change $\varepsilon = 10^{-2}$, $\mu(t)$ is ramped linearly from $\mu = -1$ to $\mu = 0$ at rate ε . With an initial condition at base the right hand potential well, the evolution of the PDF $p(x, t)$ is computed numerically using Matlab's `pdepe` function. (a) Shows the probability $p_R(t)$ of remaining in the right hand potential well over time as $\mu(t)$ is ramped. (b) Shows the evolution of the numerical solution of the PDF $p(x, t)$.

In Fig. 6.9 for each each a numerical solution of the FPE we present $p_R(t)$. The line plotted in Fig. 6.8a can also be see for the row at $D = 10^{-1}$ on the panel for $\varepsilon = 10^{-1}$ in Fig. 6.9. In Figs. 6.10 and 6.11, errors of $p_R(t)$ approximated with a 4-mode (3 non-zero eigenmodes) RSPM and 2-mode (1 non-zero eigenmode) BCPM are presented. The 4-mode RSPM directly applies Eq. (4.10) in an iterative Euler scheme. In contrast the 2-mode BCPM assumes an

adiabatic regime, and thus two coefficients are given by,

$$\frac{da_0}{dt} = 0, \quad \frac{da_1}{dt} = a_1(t)\lambda_1,$$

where the approximated PDF $\hat{p}(x, t)$ is given by,

$$\hat{p}(x, t) = a_0(t)\psi_0(x) + a_1(t)\psi_1(x).$$

Note that unlike similar looking plots presented in Secs. 6.2 and 6.3, for Figs. 6.9 to 6.11 $\mu(t)$ is ramped and thus each row is a single time-dependent solution (or errors associate with a single time-dependent solution).

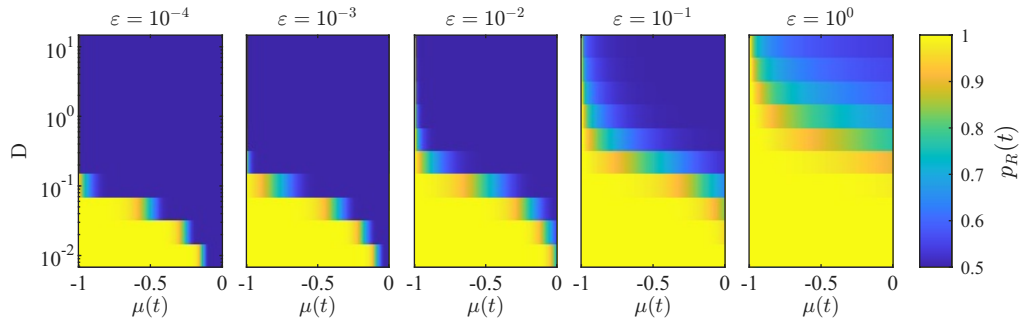


Figure 6.9: For a each diffusion constant D and parameter rate of change ϵ , with an initial condition at base the right hand potential well, probability $p_R(t)$ of remaining in the right hand potential well over time ($\mu(t)$ is ramped linearly from $\mu = -1$ to $\mu = 0$ at rate ϵ) is computed numerically using Matlab's `pdepe` function.

The 4-mode RSPM approximation does not appear to perform as well as desired in time-dependent approximations of the PDF of FPE. Naturally, for low noise regimes this result is expected as the RSPM performs poorly at approximating the eigenvalues and eigenfunctions in low noise regimes. However, in high noise settings, perhaps this was not as expected. Although the 4-mode RSPM appears to perform well for high noise regimes with low parameter rates of change ϵ in Fig. 6.10, in these settings the PDF rapidly decays to the steady state distribution over a relatively short portion of the simulation. Therefore for almost the entirety of the simulation $p_R(t)$ has decayed to $p_R(t) \approx 0.5$ (see Fig. 6.9) which is not difficult for our approximation method to follow.

In contrast, for higher parameter rates of change ϵ , the RSPM starts to struggle. We are particularly interested accurate approximations of the regions of Fig. 6.9 which transition from $p_R = 1$ to $p_R = 0.5$, prior to this transition region almost no tipping is taking and afterwards there is sufficiently large tipping rates that the system is in equilibrium. In these transition regions there does not appear to be high levels of accuracy. There are several factors which could contribute to the low accuracy in these regions:

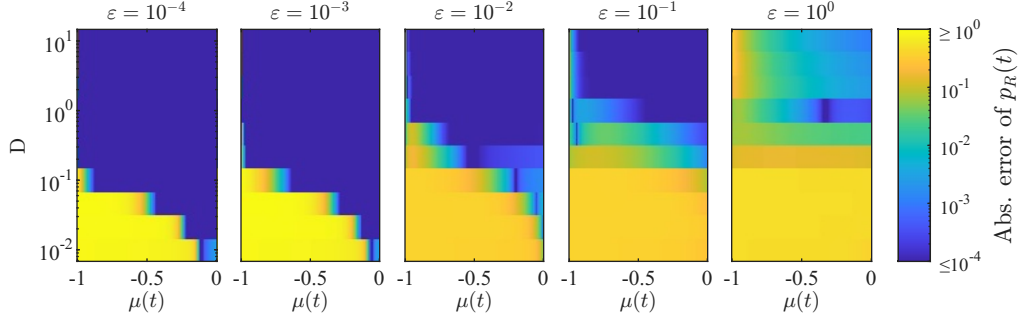


Figure 6.10: Absolute relative errors when using a 4-mode (3 non-zero eigenmodes) RSPM approximation of the probability $p_R(t)$ of remaining in the right hand potential well over time, given an initial condition at base the right hand potential well. For a each diffusion constant D and parameter rate of change ε , $\mu(t)$ is ramped linearly from $\mu = -1$ to $\mu = 0$ at rate ε .

- In high noise settings, there is a very small spectral gap between the leading non-zero eigenvalue and higher eigenvalues. As a result, a large number of eigenmodes persist over time at ‘similar’ time scales. Therefore although the 4-mode RSPM has access to 3 non-zero eigenmodes, this may not be sufficient to approximate the system at hand in which many eigenmodes may persist with non-negligible contributions. This is particularly the case for high ε values.
- For a given diffusion constant D , the RSPM is weakest for very low bifurcation parameters $\mu(t)$. However the simulations start with the lowest bifurcation parameters $\mu(0) = -1$ and then ramp $\mu(t)$ towards more favourable values for the RSPM. This could have the effect of introducing an error to the approximations of the 4-mode RSPM early in the simulation, which is carried through the remainder of the simulation.
- The FPO has an inner product which evolves over time as the potential function U evolves. Consequently, the normalisation of eigenfunctions and evolves with time as well. This adds another temporal effect which the basis coefficients must account for. In addition, as seen in Eq. (4.10), the evolution for a given basis coefficient a_n is dependent on every basis coefficient a_i and the time derivative of every eigenfunction ψ_i for $i = 0$ to $i = \infty$. That is to say, the truncation error by only tracking 4 eigenmodes is two-fold, we miss out on the ‘description’ of the system provided by truncated modes, and we also see increased errors in the modes we do include.

In general, the results are very promising for the 2-mode BCPM approximation. As seen in Fig. 6.11, for $\varepsilon = 10^{-4}$, $\varepsilon = 10^{-3}$ and even $\varepsilon = 10^{-2}$, the errors of the 2-mode BCPM are very low. The transition regions start to introduce small errors for $\varepsilon = 10^{-2}$, which expands to beyond the transition regions for $\varepsilon = 10^{-1}$, $\varepsilon = 10^0$. These errors become particularly evident

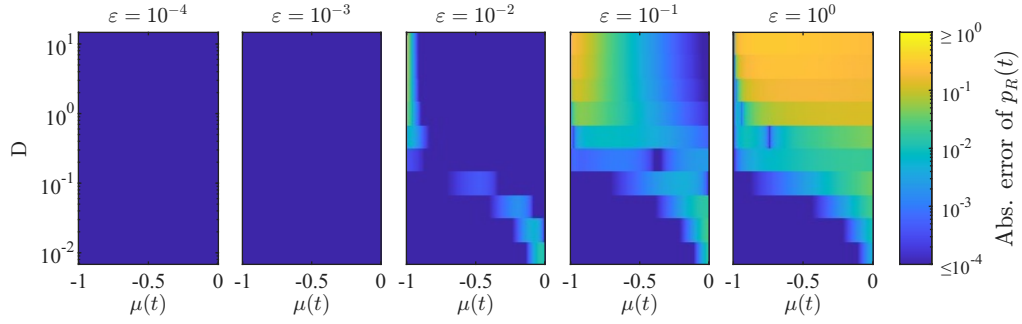


Figure 6.11: Absolute relative errors when using a 2-mode (1 non-zero eigenmode) BCPM approximation of the probability $p_R(t)$ of remaining in the right hand potential well over time, given an initial condition at base the right hand potential well. For a each diffusion constant D and parameter rate of change ε , $\mu(t)$ is ramped linearly from $\mu = -1$ to $\mu = 0$ at rate ε .

for high diffusion constants D with high parameter rates of change ε , which is expected given the regions for which the BCPM provides accurate eigenvalue approximations. An accurate parameter range of $\varepsilon \leq 10^{-2}$ provides very wide range of applications in real world systems.

Chapter 7

Discussion

The benefits of an eigenmode approximation are best seen in cases where a certain number of non-zero eigenmodes (typically only the leading non-zero mode) dominate the eigenmode expansion. This allows a truncation of the eigenmode expansion to be made (giving the eigenmode approximation) while still retaining high accuracy. This occurs for the FPE when there is a large potential well relative to the noise, causing the leading non-zero mode to dominate. As a result, although introducing the RSPM (to complement the BCPM) to allow for accurate eigenvalue and eigenfunction approximations in high noise settings, some of the benefits of an eigenmode approximation are lost in these settings.

As such, further extending the BCPM to allow approximations of eigenvalues and eigenfunctions for all potential functions (or at least even more forms) is a logical direction for future research. The simplicity of implementing the BCPM and its applicability to low noise settings is very appealing, however being restricted to even confining potentials is naturally a big limitation.

The intermediate region of poor eigenvalue/eigenfunction approximations seen in Fig. 6.4 is another obvious direction for future research. Not only is this region of the parameter space lacking in an accurate approximation method, but it is also of high real world importance. The region for which the BCPM performs best has a low tipping rate, much of which is so low that one can reasonably assume that no tipping event will take place in the immediate future. On the other hand, in the high noise setting which is well explained by the RSPM, there is such a high probability of a tipping event taking place in the near future that it is almost not worth trying to take action. The intermediate region however, is the region in which a tipping event is of serious immediate threat (i.e a tipping rate between $r = 0.2$ and $r = 0.5$) yet it is not guaranteed that tipping will take place. This is the region for which a timely intervention in a real world system could have huge risk mitigating effects. Consider the example of over exploited fisheries, intervention (i.e. lowered catch quotes, stricter regulations) is of highest importance when there is a serious risk of tipping (i.e. fish population collapse), but while

there is still time for such interventions to be implemented and take effect prior to tipping. That being said, it seems difficult to estimate eigenvalues and eigenfunctions without a clear ‘drift-dominate’ regime, or ‘noise-dominate’ regime to base the approximations off.

There are also opportunities to further explore the implementation of the RSPM. In this thesis, the RSPM presented in Sec. 5.3 was applied to make perturbations to the FPO. As discussed in Sec. 3.4, the equivalence between the Schrödinger operator and the FPO means that the eigenvalues between the two operators are the same, and a transformation can be applied to find corresponding eigenfunctions. Therefore we could instead apply the RSPM to the Schrödinger operator directly. After finding the approximated eigenvalues and eigenfunctions of the corresponding Schrödinger operator, one could then transform them into the Fokker-Planck eigenfunctions. A reason why we might want to explore such an approach is that the Schrödinger operator does not have a weighted inner-product. As such, perturbing functions can be done so without the complications of an evolving weight function.

Chapter 8

Conclusion

In this thesis we have explored two methods for approximating the eigenvalues and eigenfunctions of the FPO, which were then applied to eigenmode approximations of tipping point systems. Specifically, we focused on noisy bifurcation tipping. We expanded upon the boundary condition (BC) perturbation method presented by Ritchie and Sieber [22] by allowing the even confining potential function, and even anti-confining potential function to be approximated. We also applied the RSPM, a classical method in quantum mechanics, to approximate the eigenvalues and eigenfunctions of the FPO. The two methods were then applied to an even bistable confining potential function across a range of parameters. When approximating the eigenvalues and eigenfunctions of the FPO the two methods complemented one another, as they each provided accurate approximations for different regions of the parameter space. The BCPM performed best in settings with low diffusion relative to the depth of the potential wells, whereas the RSPM performed best in settings with high diffusion relative to the depth of the potential wells. In a time-dependent setting with potential functions approaching a bifurcation, eigenmode approximations were compared. The BCPM performed well for slowly changing systems. The RSPM performed best in high diffusion settings but often could not follow rapidly evolving systems.

References

- [1] Ashwin, P., Wieczorek, S., Vitolo, R., Cox, P.: Tipping points in open systems: bifurcation, noise-induced and rate-dependent examples in the climate system. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* **370**(1962), 1166–1184 (2012)
- [2] Beddington, J.R., May, R.M.: Harvesting natural populations in a randomly fluctuating environment. *Science* **197**(4302), 463–465 (1977)
- [3] Bender, C.M., Orszag, S.A.: Advanced mathematical methods for scientists and engineers: Asymptotic methods and perturbation theory, vol. 1. Springer (1999)
- [4] Bennewitz, C., Weikard, R., Brown, M.: Spectral and scattering theory for ordinary differential equations. Springer (2020)
- [5] Berglund, N., Gentz, B.: Noise-induced phenomena in slow-fast dynamical systems: a sample-paths approach. Springer (2006)
- [6] Broecker, W.S.: Unpleasant surprises in the greenhouse? *Nature* **328**(6126), 123–126 (1987)
- [7] Dakos, V., Matthews, B., Hendry, A.P., Levine, J., Loeuille, N., Norberg, J., Nosil, P., Scheffer, M., De Meester, L.: Ecosystem tipping points in an evolving world. *Nature ecology & evolution* **3**(3), 355–362 (2019)
- [8] Dakos, V., Scheffer, M., Van Nes, E.H., Brovkin, V., Petoukhov, V., Held, H.: Slowing down as an early warning signal for abrupt climate change. *Proceedings of the National Academy of Sciences* **105**(38), 14308–14312 (2008)
- [9] Donovan, G.M.: Characterising early warning signals for critical transitions using a distribution moment approximation for the fokker–planck equation. *Physica A: Statistical Mechanics and its Applications* **646**, 129868 (2024)
- [10] Driscoll, T.A., Hale, N., Trefethen, L.N.: Chebfun guide (2014)

- [11] Gardiner, C.W., et al.: Handbook of stochastic methods, vol. 3. springer Berlin (2004)
- [12] Inc., T.M.: Matlab version: 25.2.0.3177638 (r2025b) (2025), <https://www.mathworks.com>
- [13] Kato, T.: On the adiabatic theorem of quantum mechanics. Journal of the Physical Society of Japan **5**(6), 435–439 (1950)
- [14] Kloeden, P.E., Pearson, R.: The numerical solution of stochastic differential equations. The ANZIAM Journal **20**(1), 8–12 (1977)
- [15] Kramers, H.A.: Brownian motion in a field of force and the diffusion model of chemical reactions. physica **7**(4), 284–304 (1940)
- [16] Kuehn, C.: A mathematical framework for critical transitions: Bifurcations, fast–slow systems and stochastic dynamics. Physica D: Nonlinear Phenomena **240**(12), 1020–1035 (2011)
- [17] Øksendal, B.: Stochastic differential equations. Springer (2003)
- [18] Rahmstorf, S.: Ocean circulation and climate during the past 120,000 years. Nature **419**(6903), 207–214 (2002)
- [19] Rahmstorf, S.: Is the atlantic overturning circulation approaching a tipping point? Oceanography **37**(3), 16–29 (2024)
- [20] Risken, H., Frank, T.: The Fokker-Planck equation: methods of solution and applications. Springer, second edn. (1992)
- [21] Ritchie, P.: Early-warning indicators for tipping points. Ph.D. thesis, University of Exeter (2016)
- [22] Ritchie, P., Sieber, J.: Probability of noise-and rate-induced tipping. Physical Review E **95**(5), 052209 (2017)
- [23] Romano, F., Kuehn, C.: Analysis and predictability of tipping points with leading-order nonlinear term. International Journal of Bifurcation and Chaos **28**(08), 1850103 (2018)
- [24] Scheffer, M., Hosper, S.H., Meijer, M.L., Moss, B., Jeppesen, E.: Alternative equilibria in shallow lakes. Trends in ecology & evolution **8**(8), 275–279 (1993)
- [25] Simon, B.: Schrodinger operators with purely discrete spectrum. arXiv preprint arXiv:0810.3275 (2008)

- [26] Skeel, R.D., Berzins, M.: A method for the spatial discretization of parabolic equations in one space variable. *SIAM journal on scientific and statistical computing* **11**(1), 1–32 (1990)
- [27] Staver, A.C., Archibald, S., Levin, S.A.: The global extent and determinants of savanna and forest as alternative biome states. *science* **334**(6053), 230–232 (2011)
- [28] Tang, K.T.: *Mathematical methods for engineers and scientists 3: Fourier analysis, partial differential equations and variational methods*. Springer (2007)
- [29] Van Westen, R.M., Kliphuis, M., Dijkstra, H.A.: Physics-based early warning signal shows that amoc is on tipping course. *Science advances* **10**(6), eadk1189 (2024)
- [30] Vellinga, M., Wood, R.A.: Global climatic impacts of a collapse of the atlantic thermohaline circulation. *Climatic change* **54**(3), 251–267 (2002)
- [31] Wissel, C.: A universal law of the characteristic return time near thresholds. *Oecologia* **65**(1), 101–107 (1984)